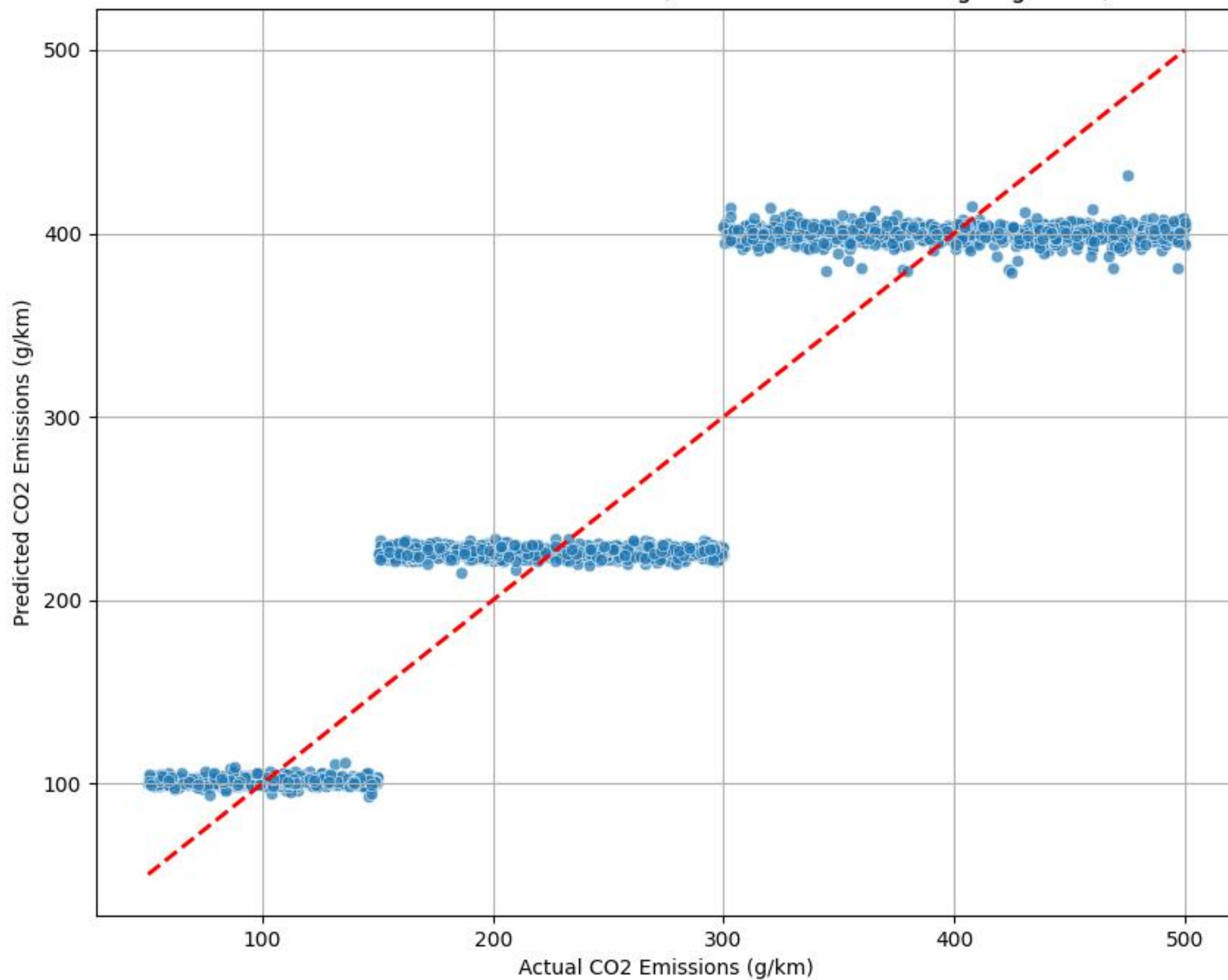
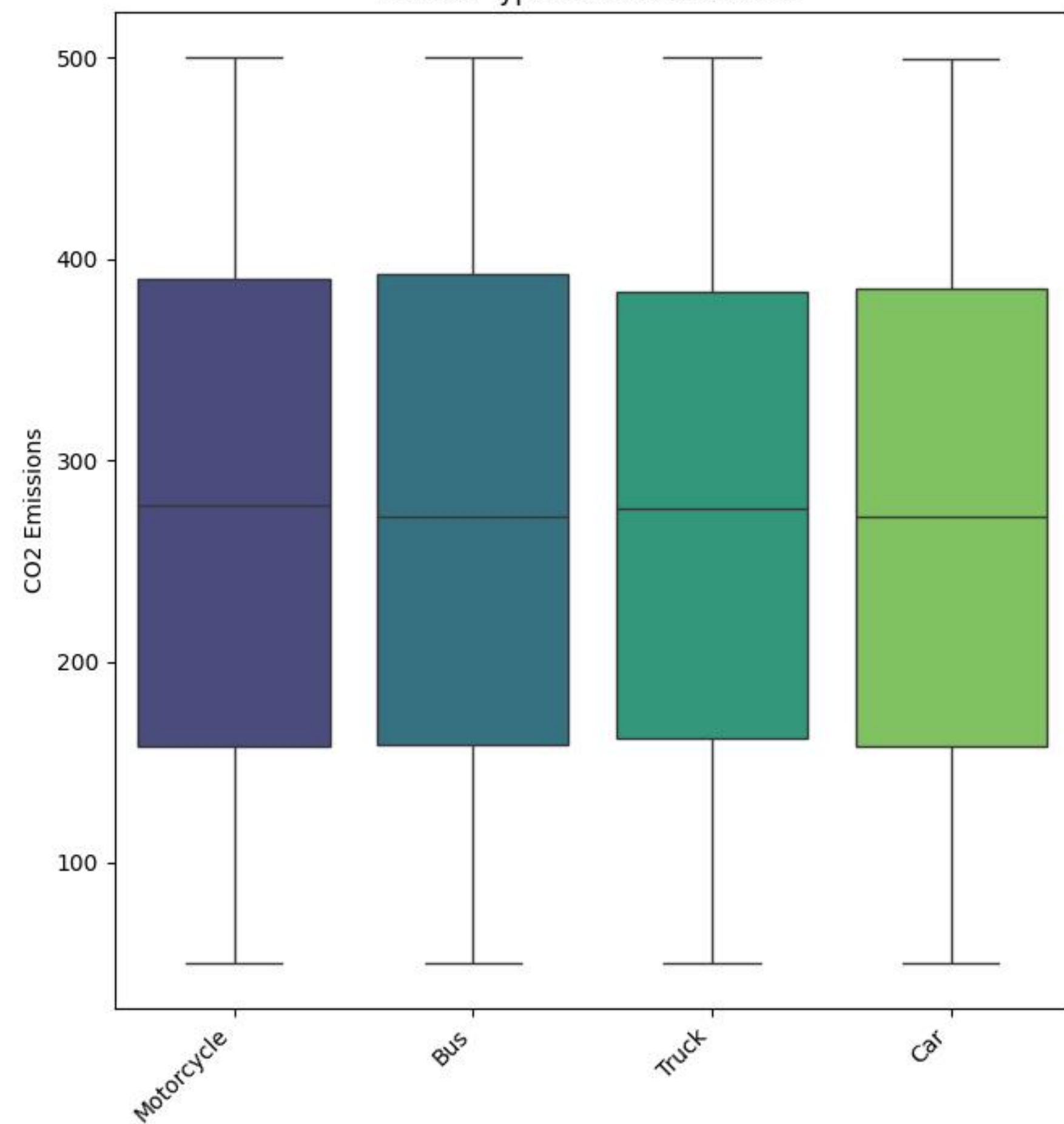


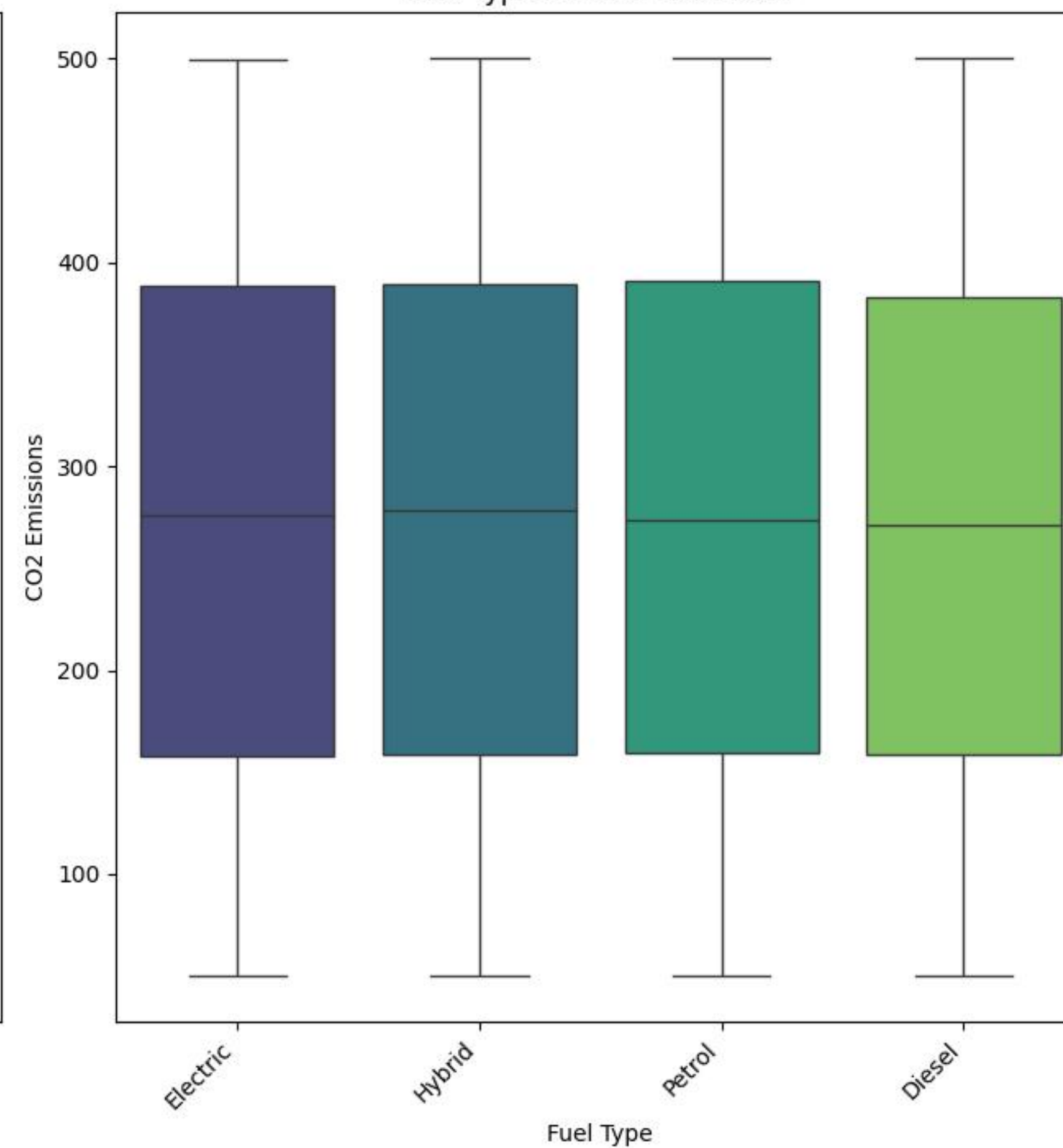
Actual vs. Predicted CO2 Emissions (Tuned Gradient Boosting Regressor)



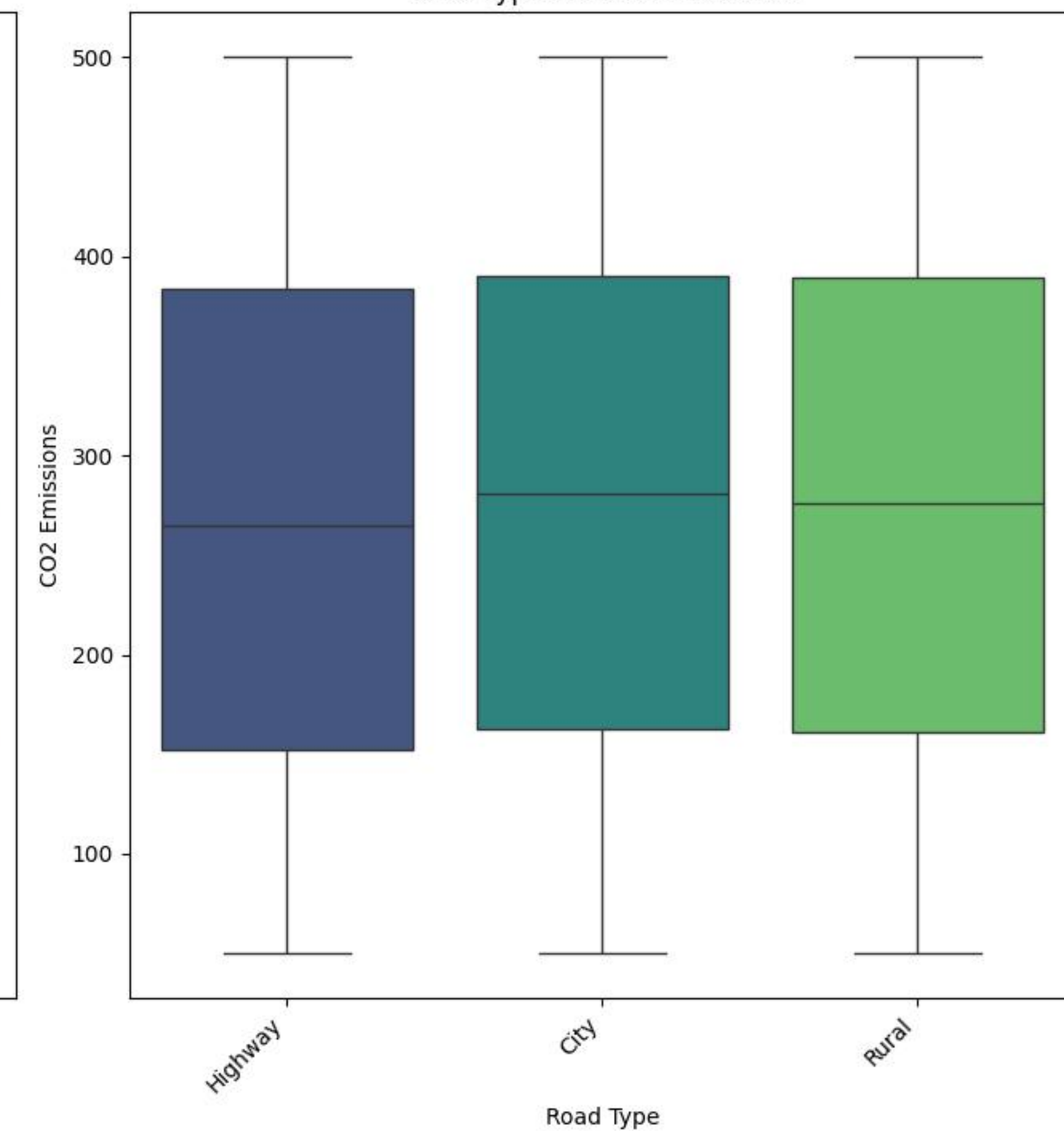
Vehicle Type vs. CO2 Emissions



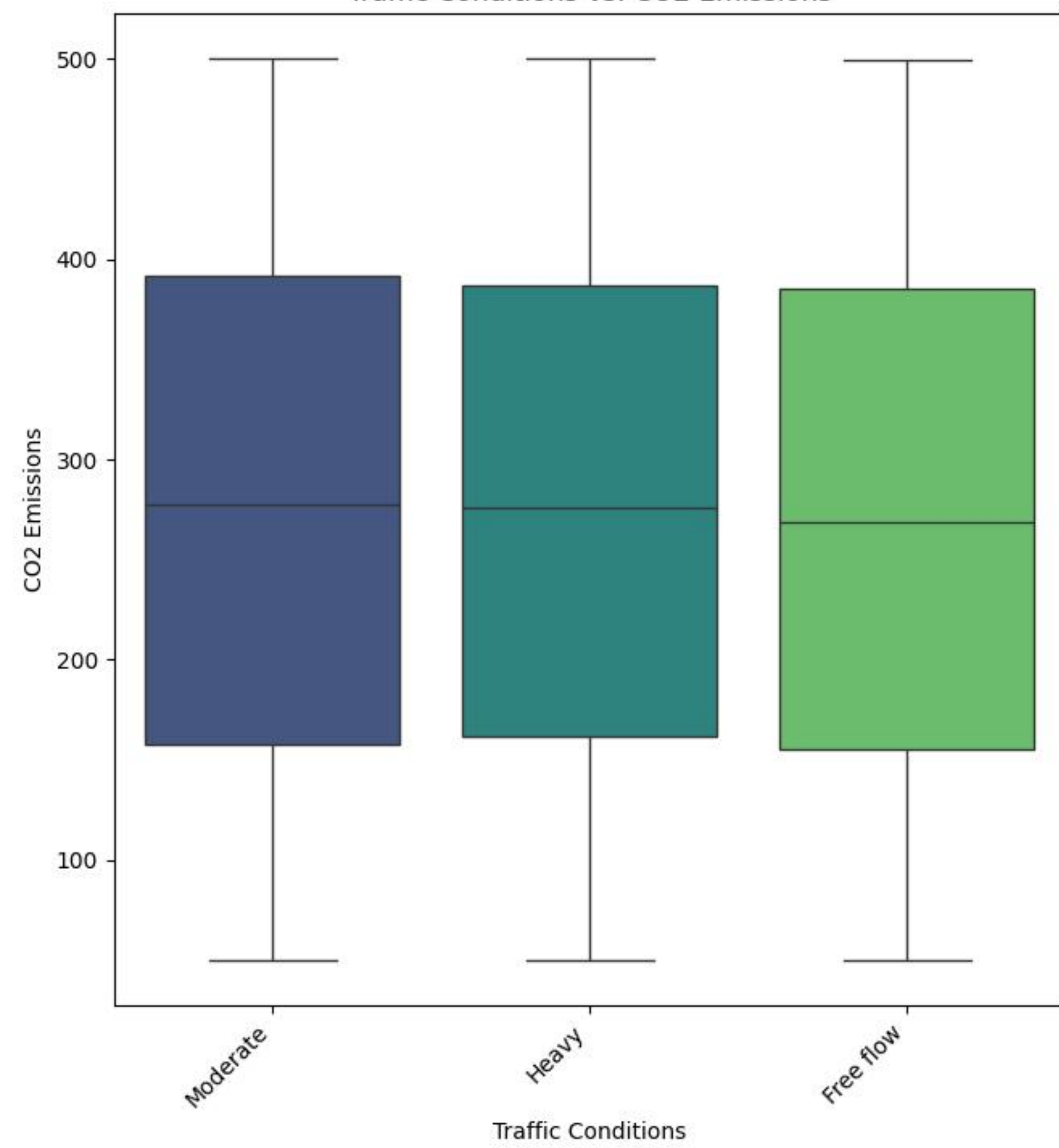
Fuel Type vs. CO2 Emissions



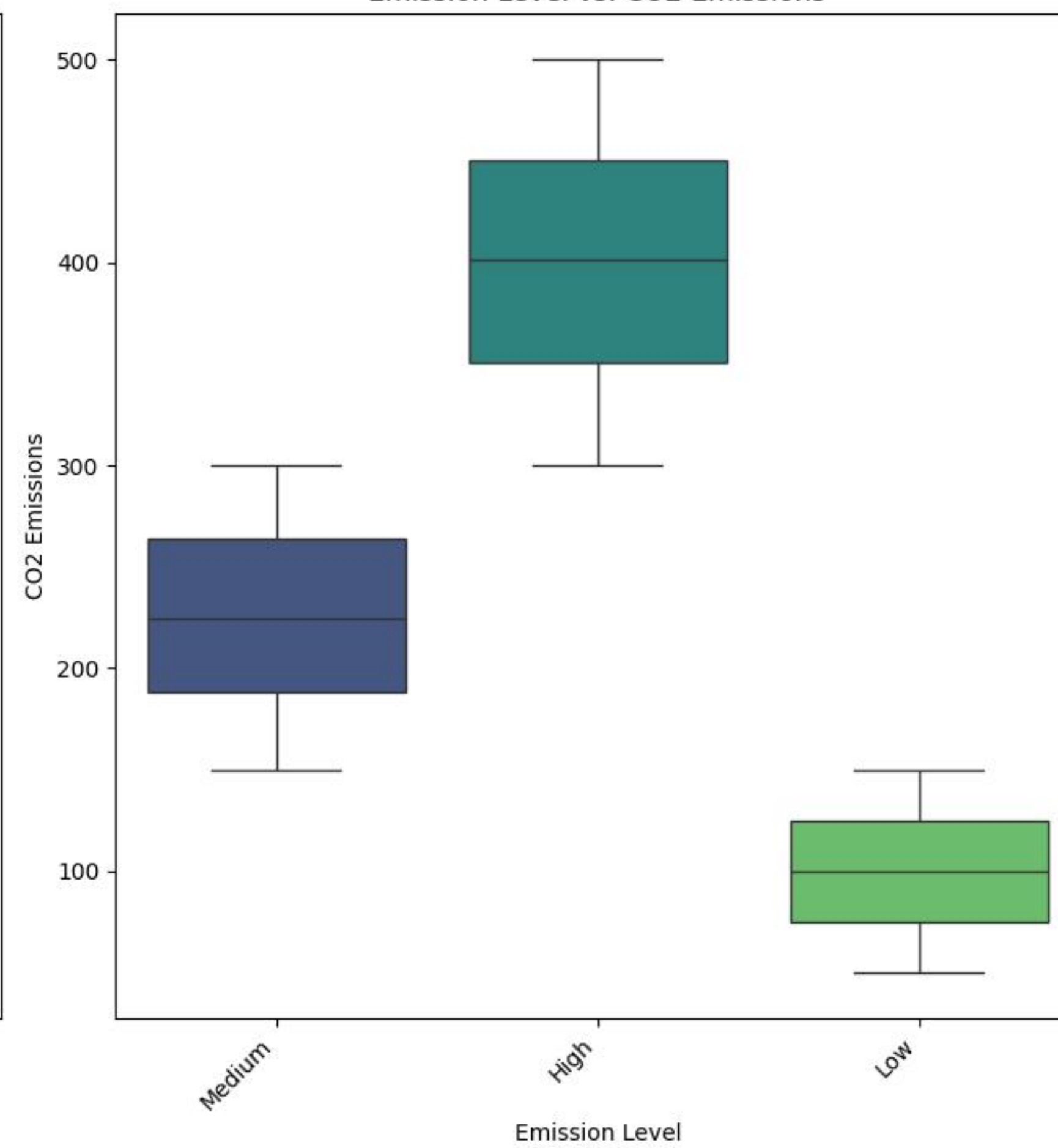
Road Type vs. CO2 Emissions



Traffic Conditions vs. CO2 Emissions

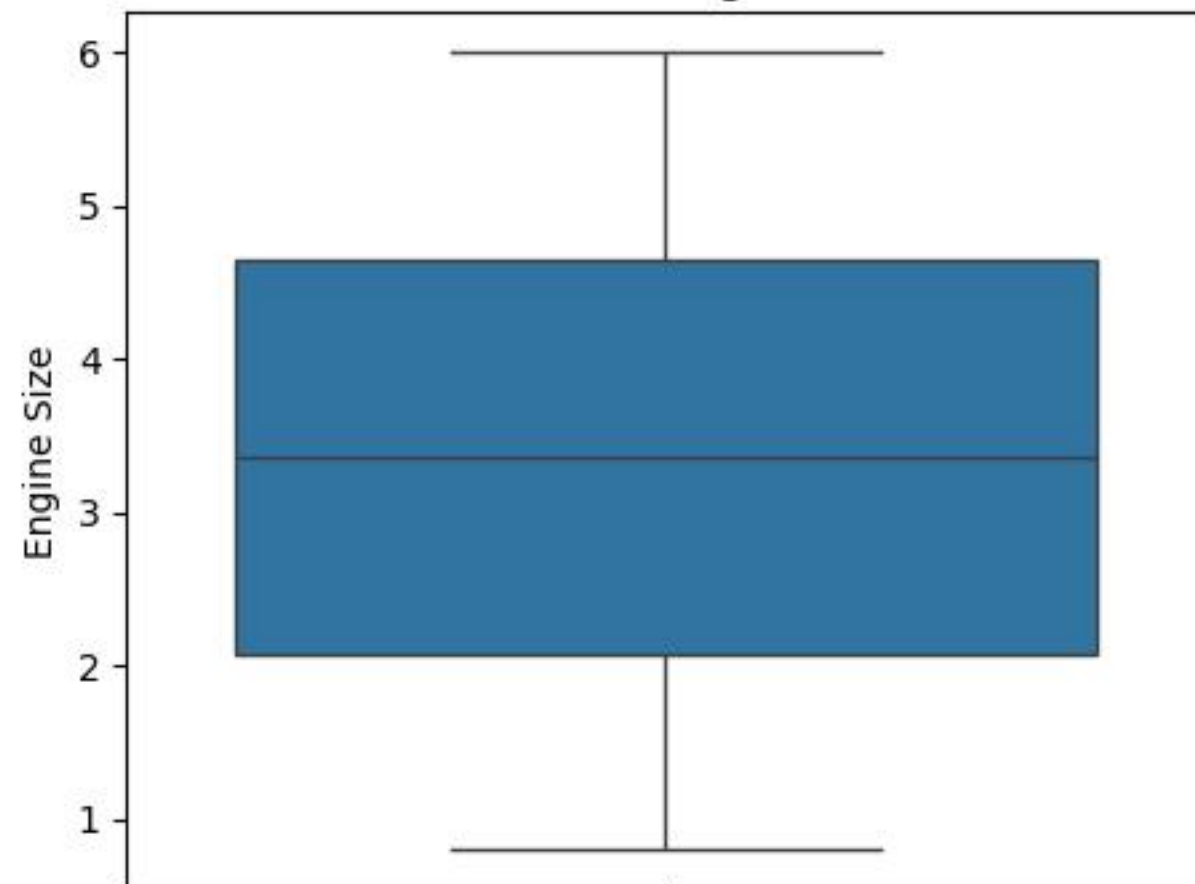


Emission Level vs. CO2 Emissions

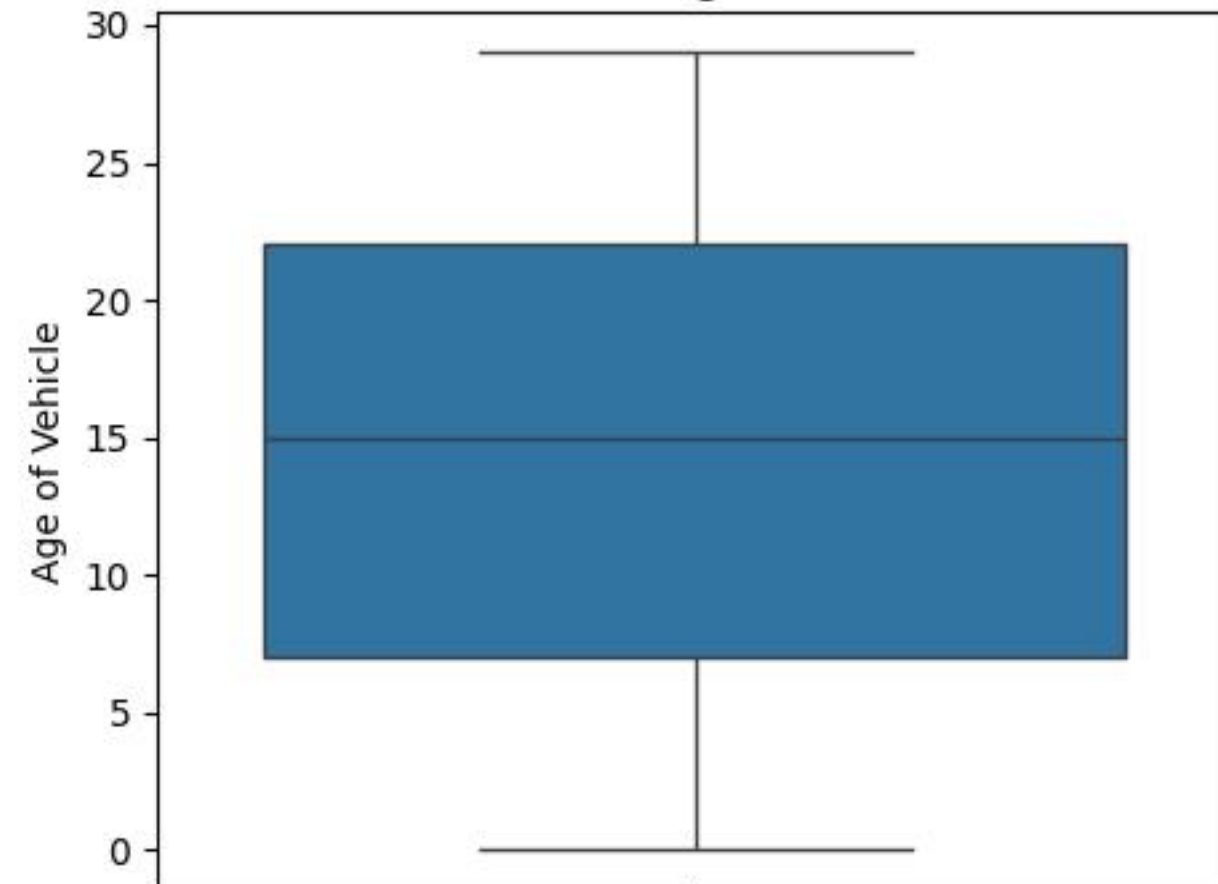




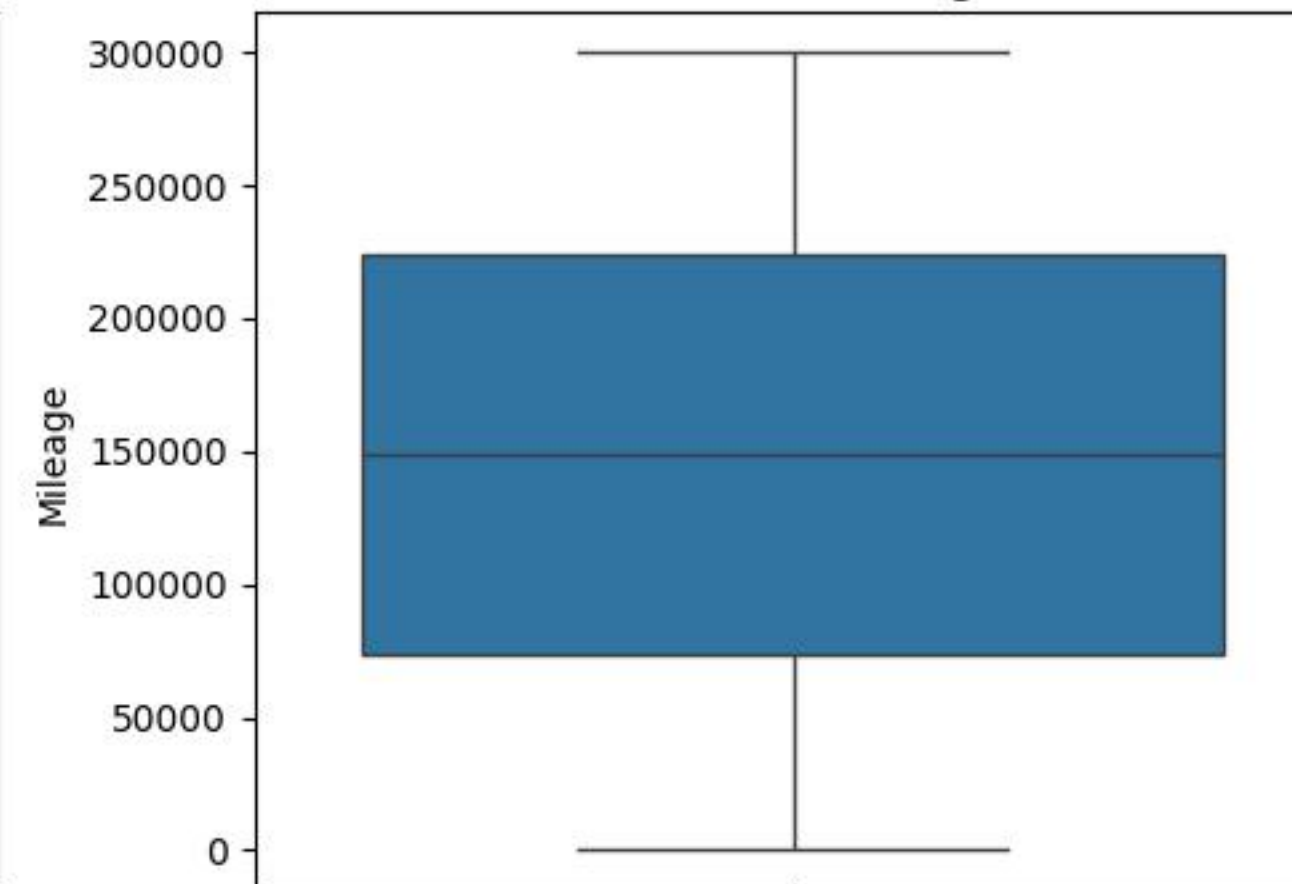
Box Plot of Engine Size



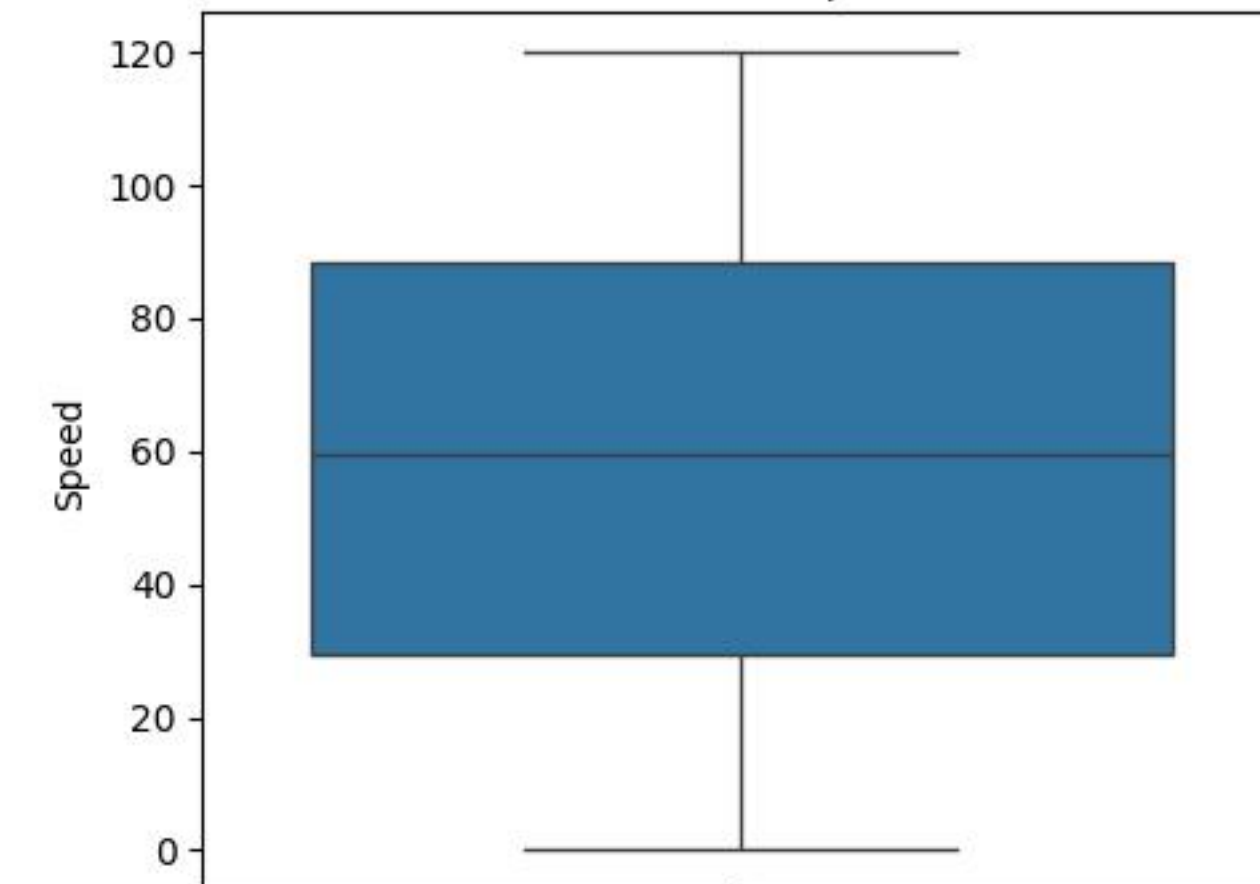
Box Plot of Age of Vehicle



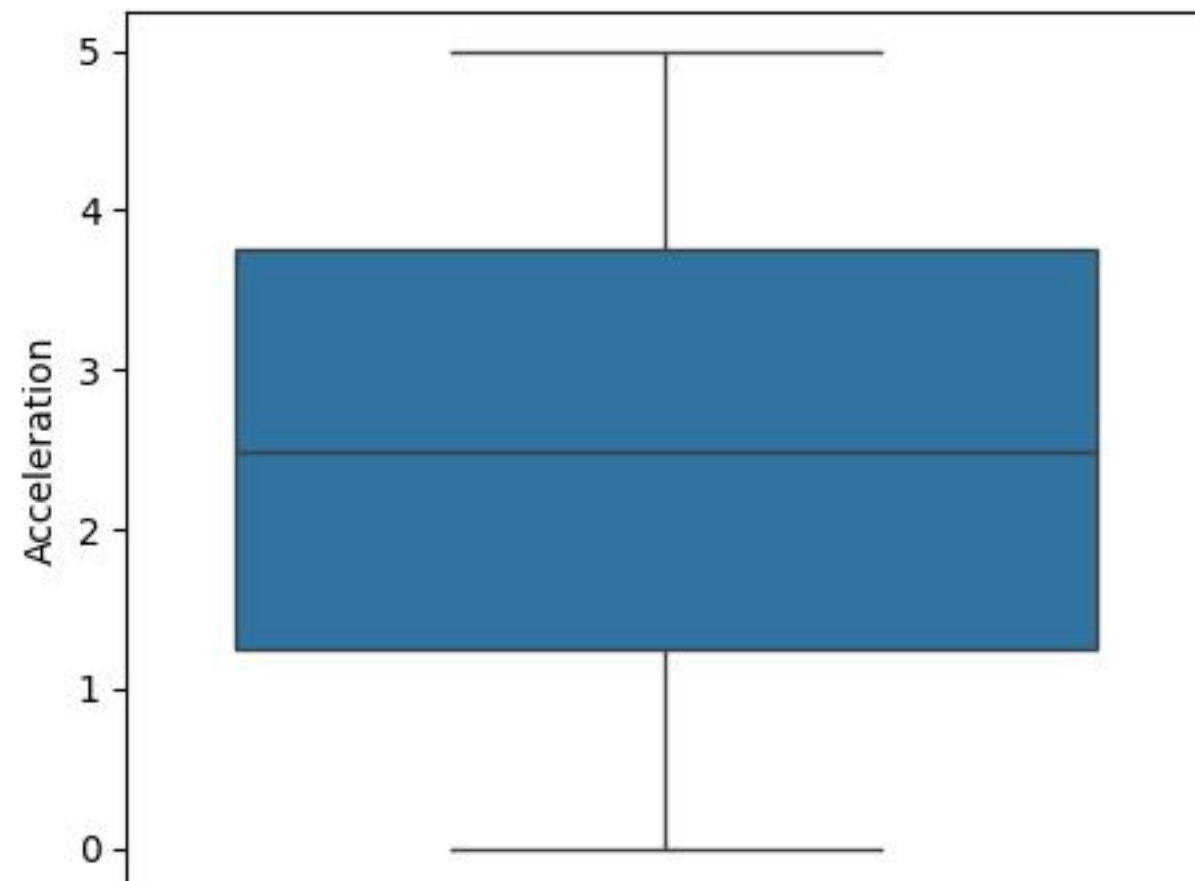
Box Plot of Mileage



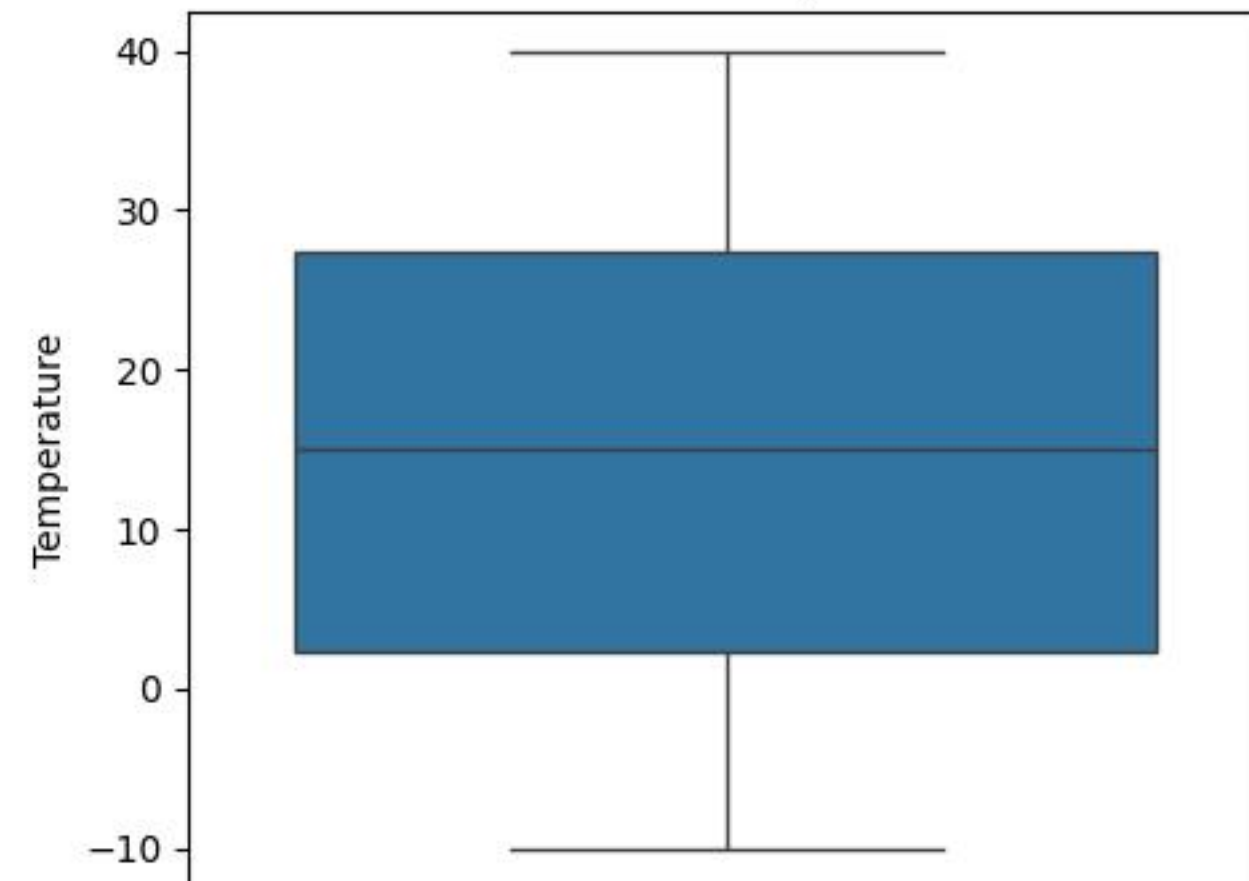
Box Plot of Speed



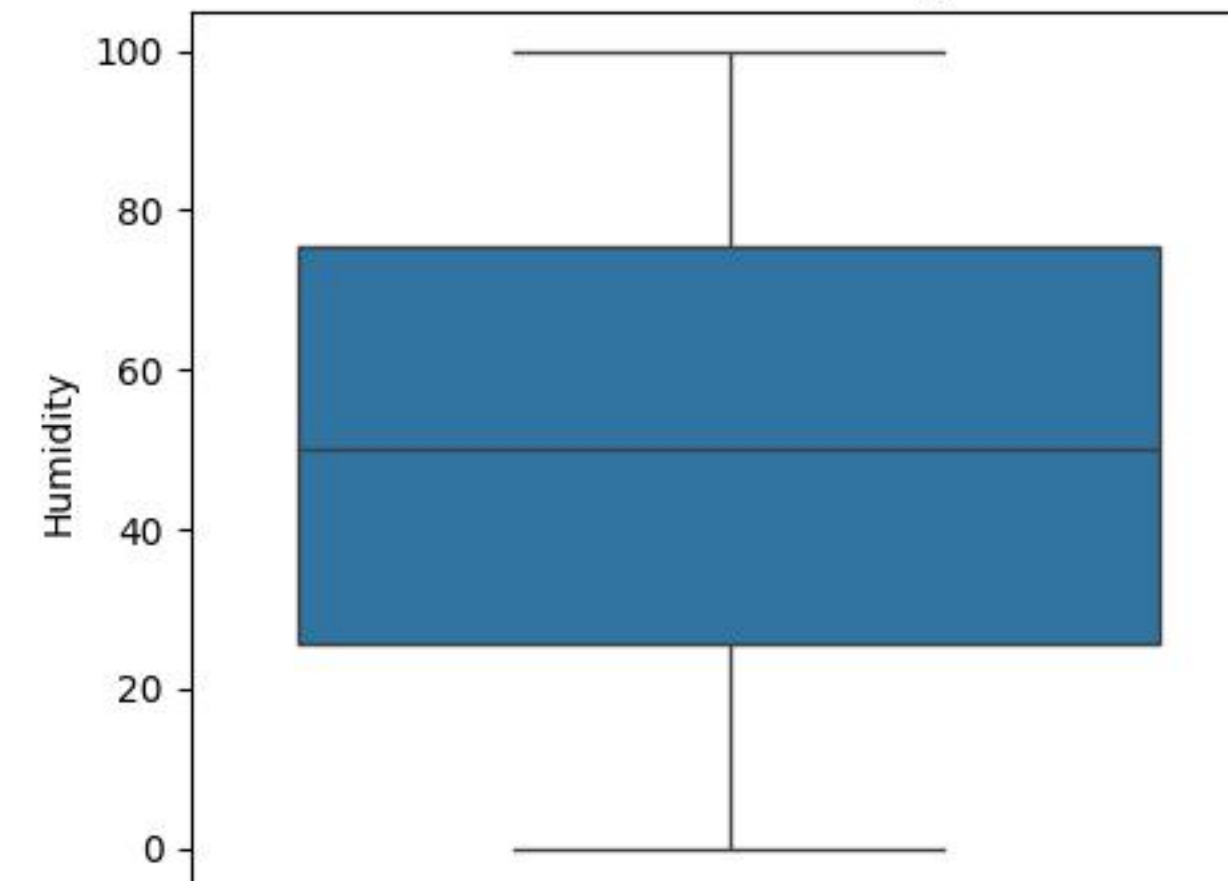
Box Plot of Acceleration



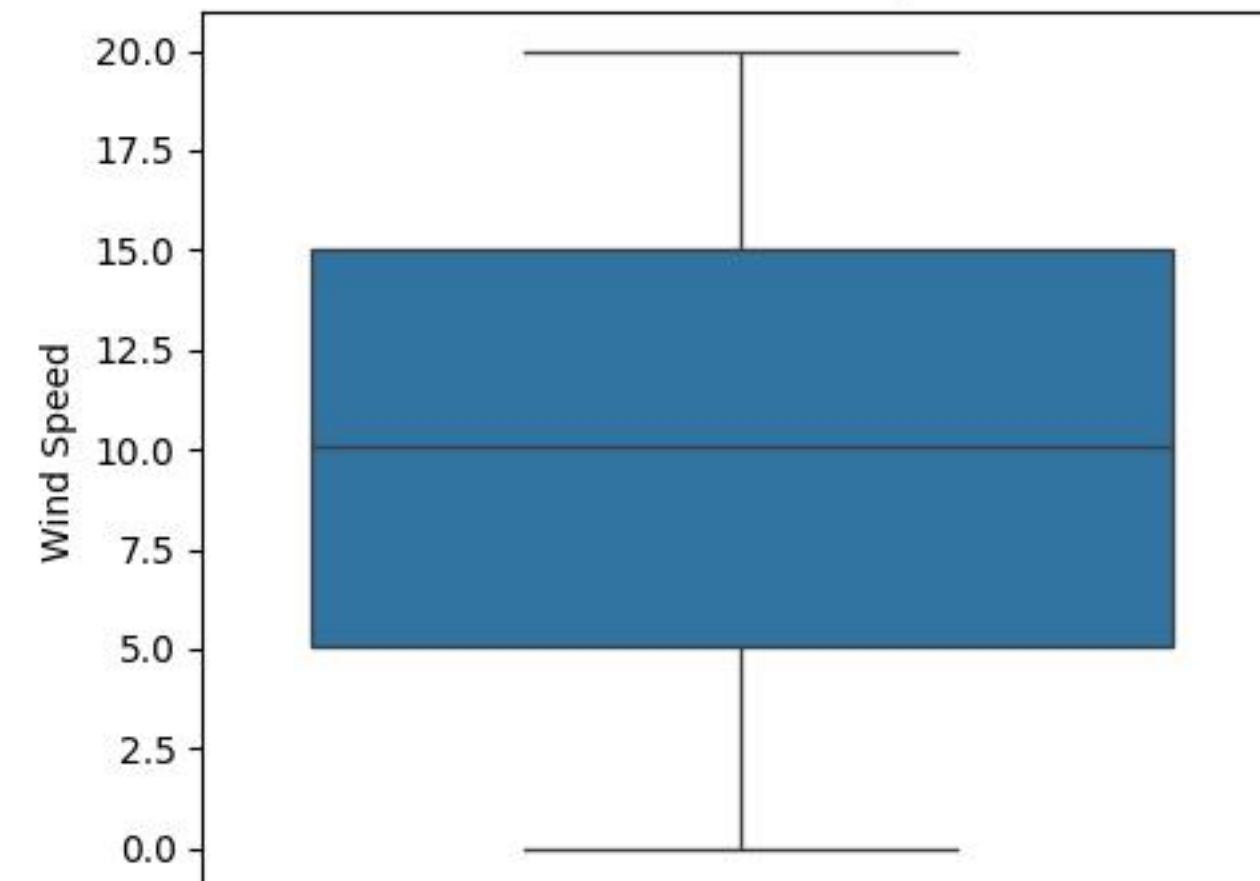
Box Plot of Temperature



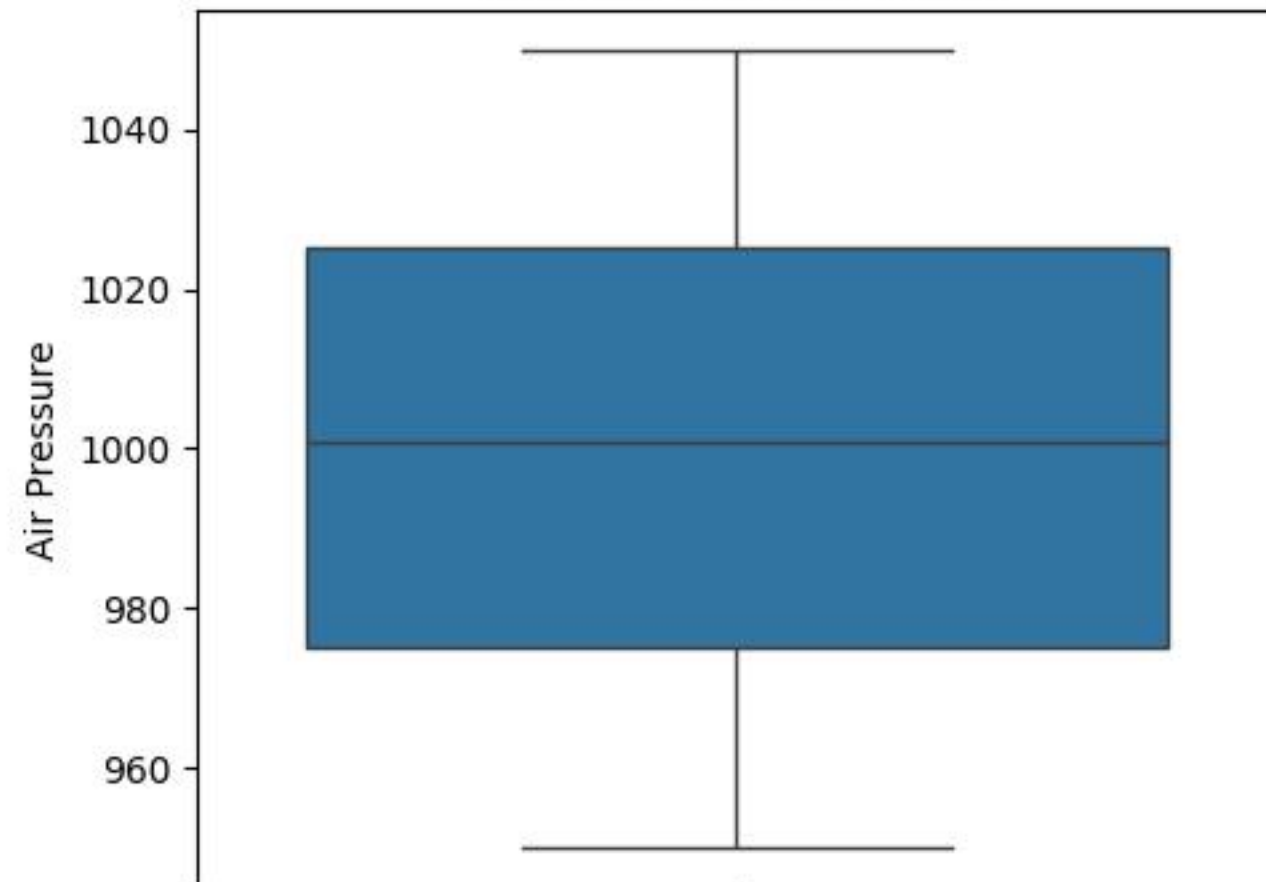
Box Plot of Humidity



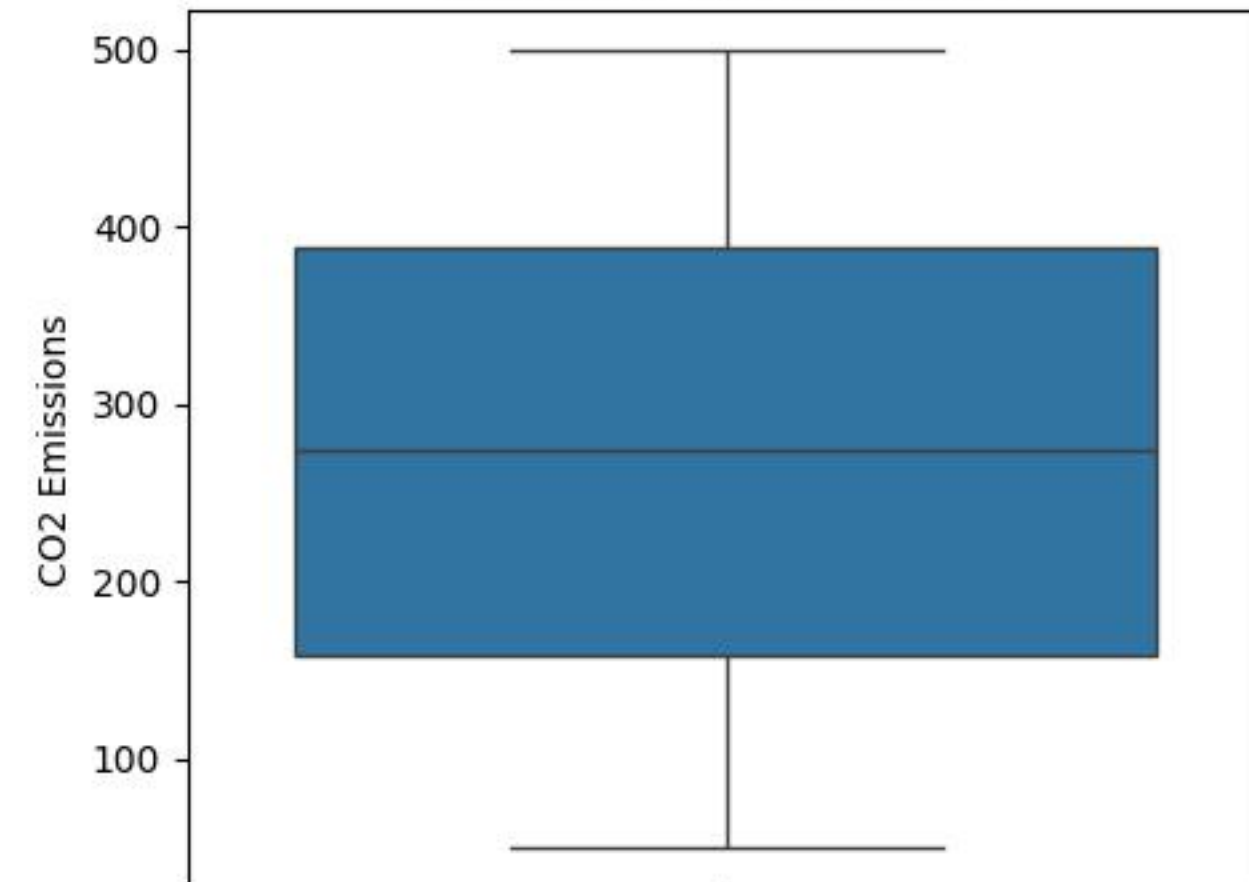
Box Plot of Wind Speed



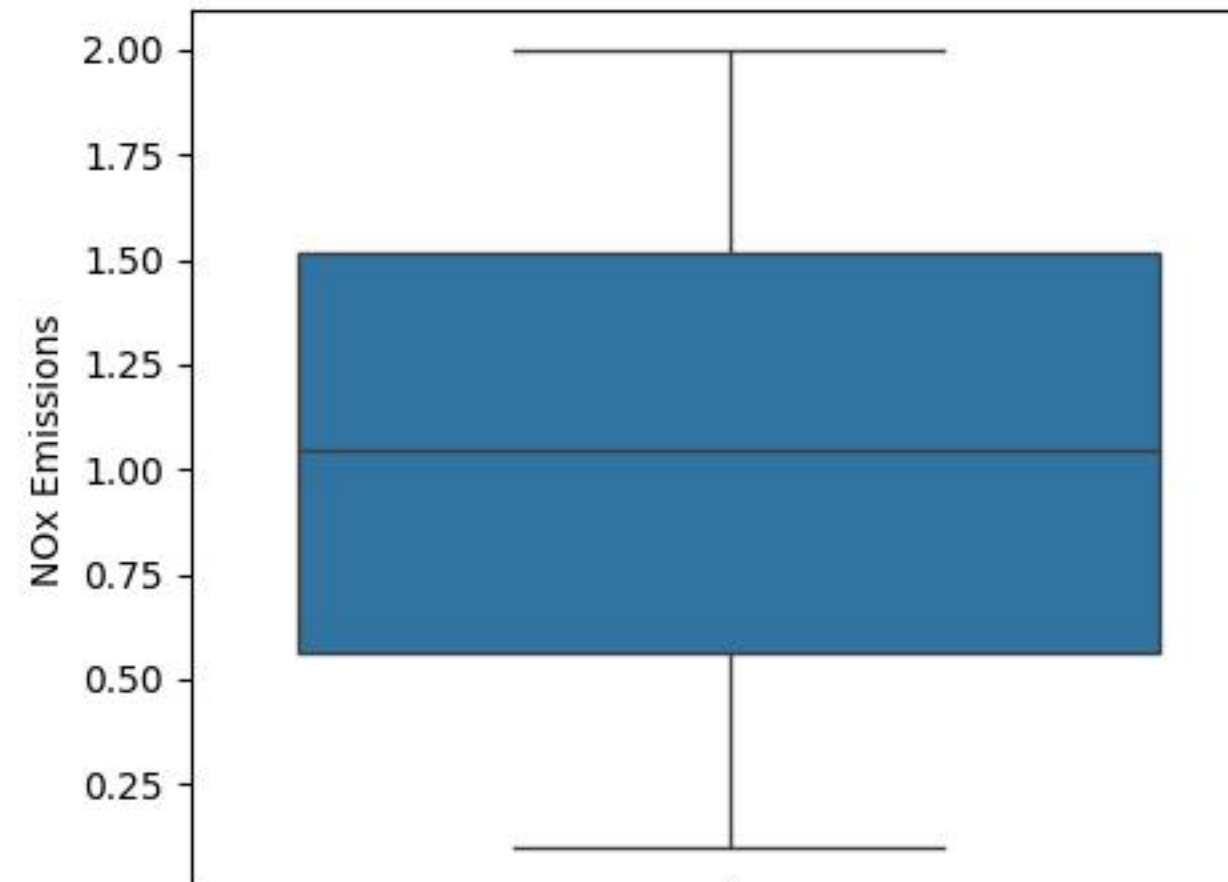
Box Plot of Air Pressure



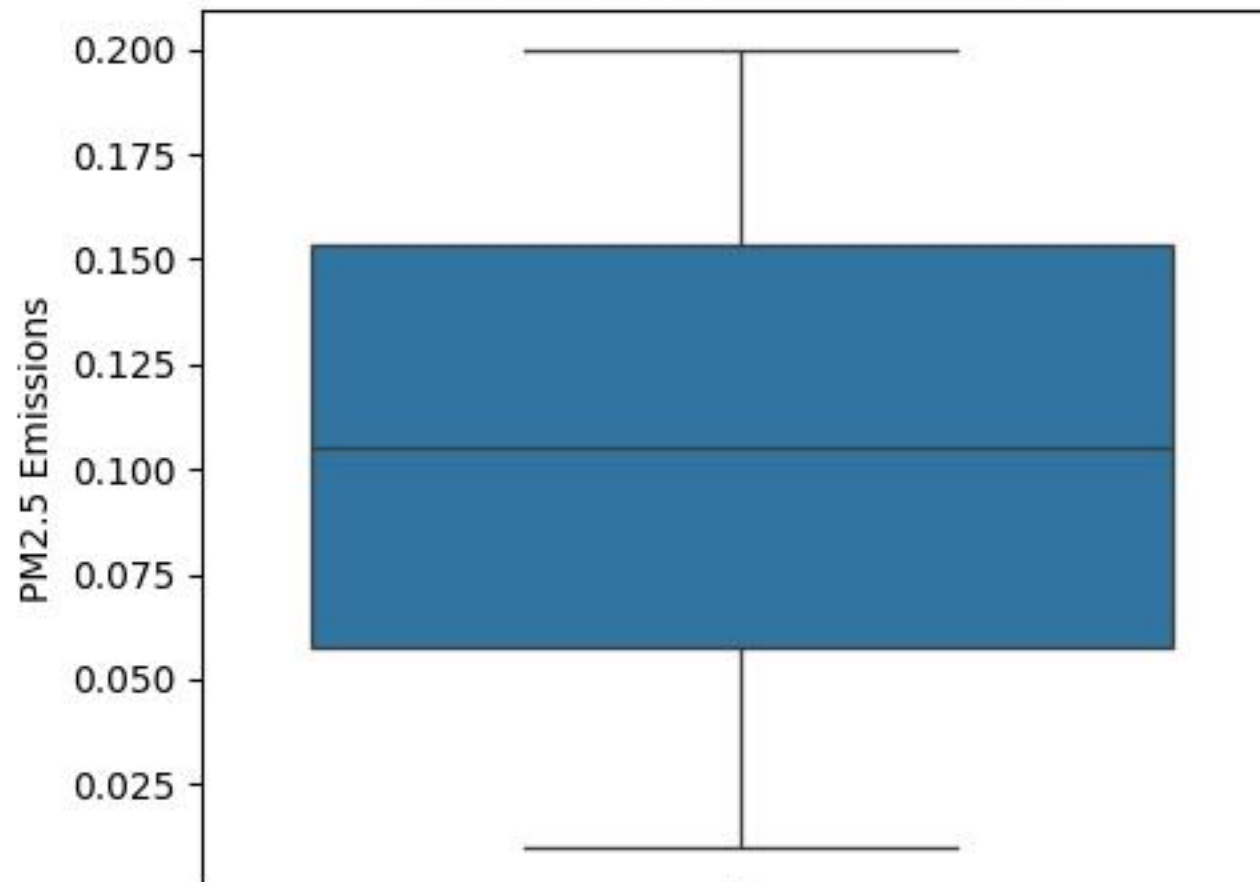
Box Plot of CO2 Emissions



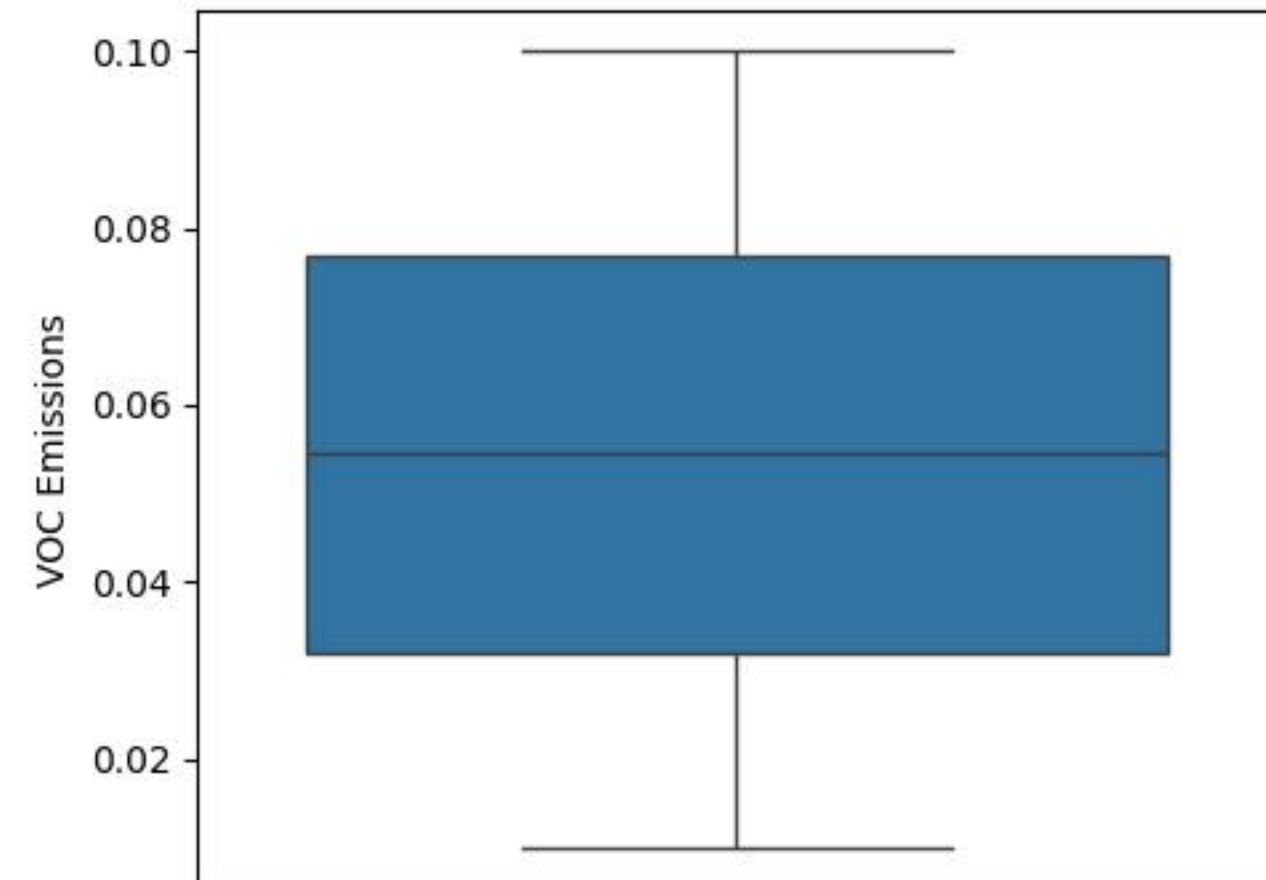
Box Plot of NOx Emissions



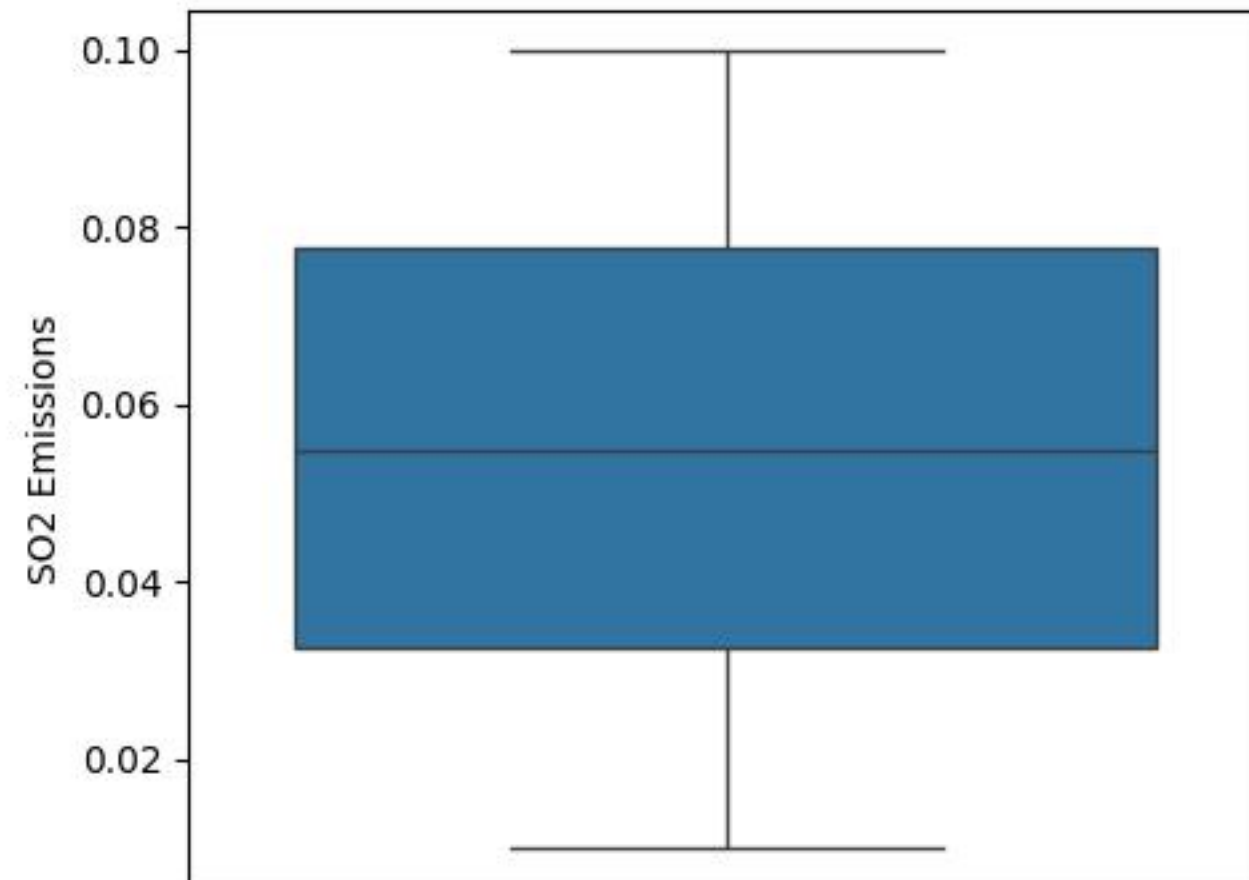
Box Plot of PM2.5 Emissions



Box Plot of VOC Emissions

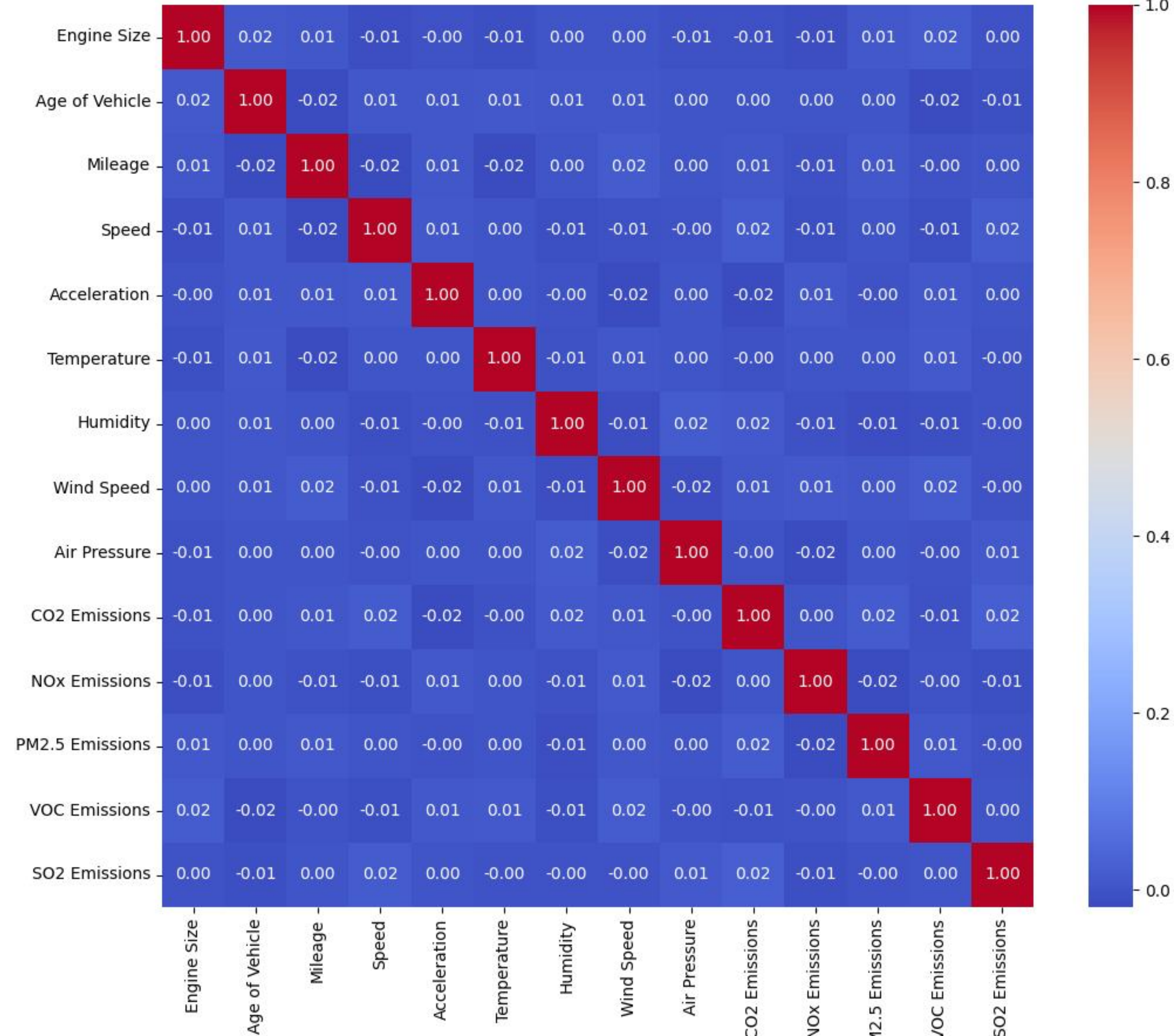


Box Plot of SO2 Emissions



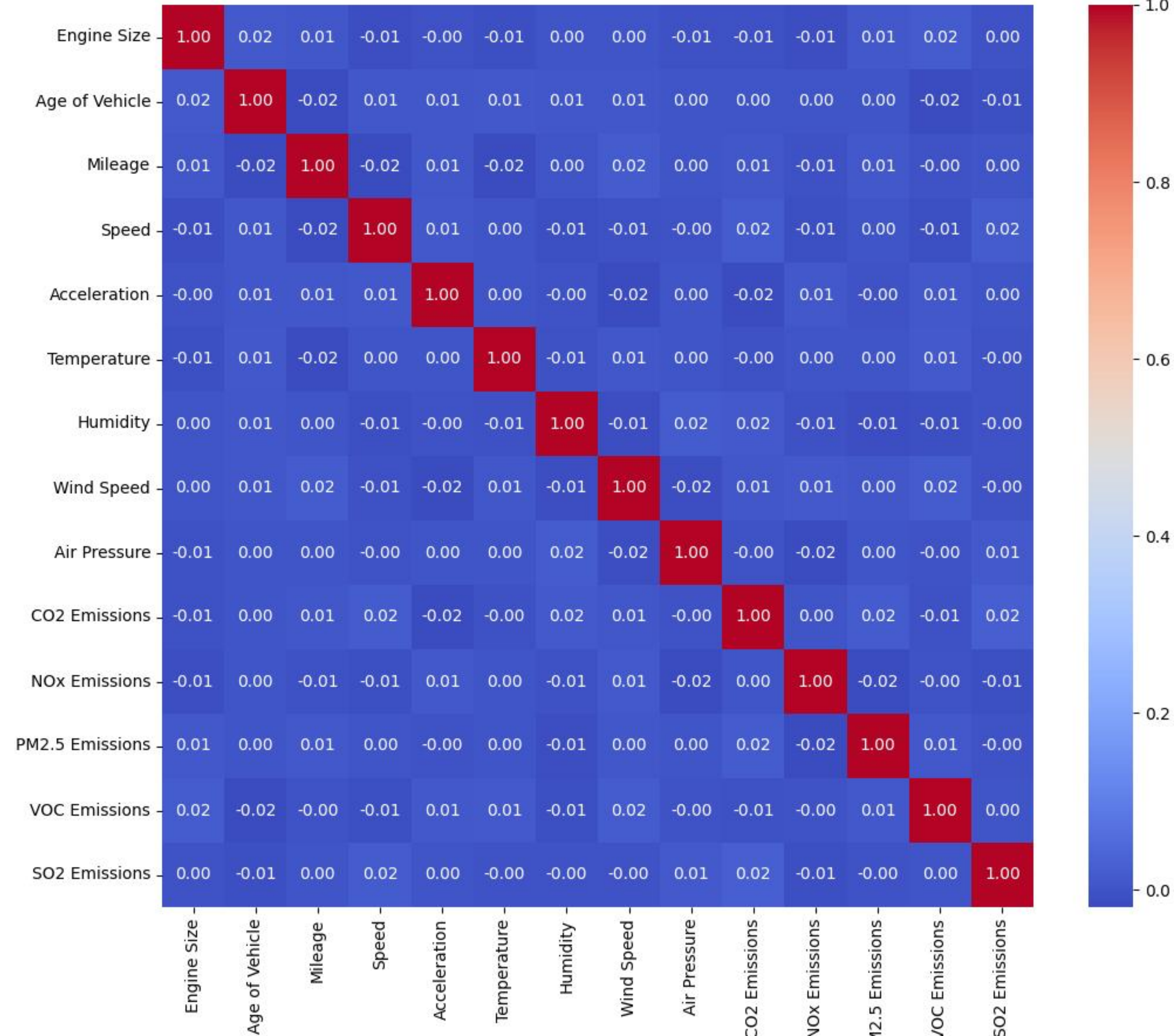


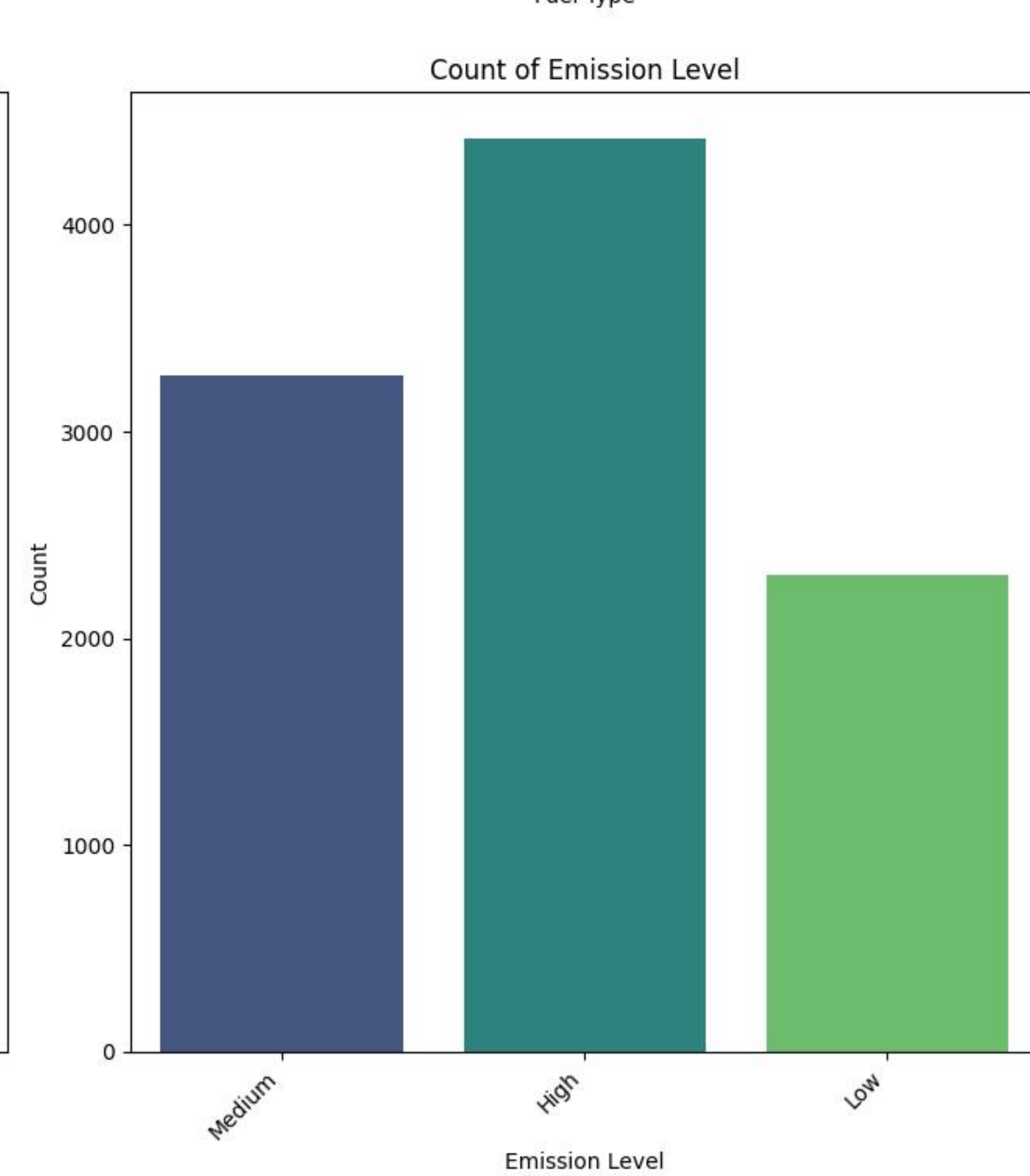
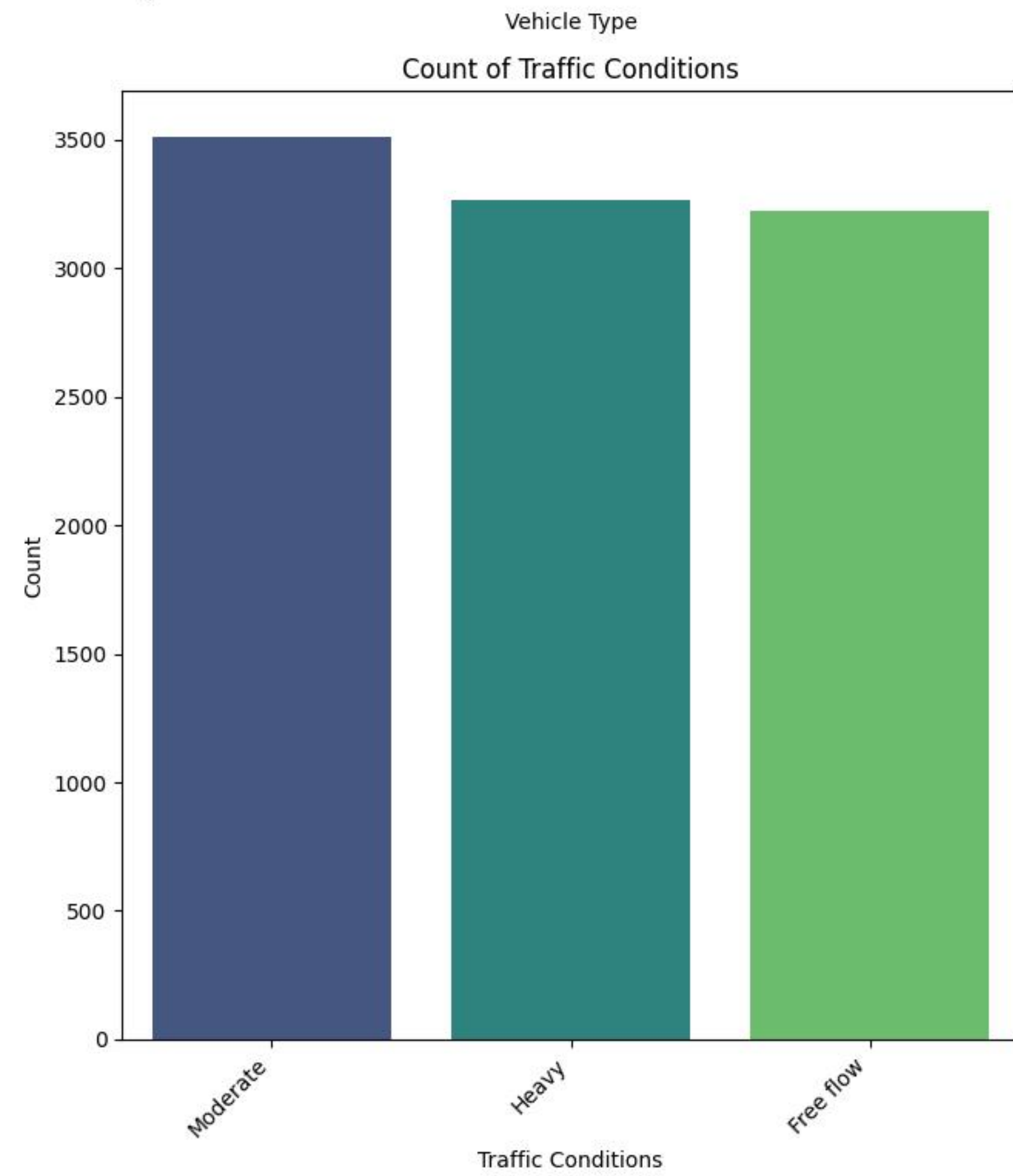
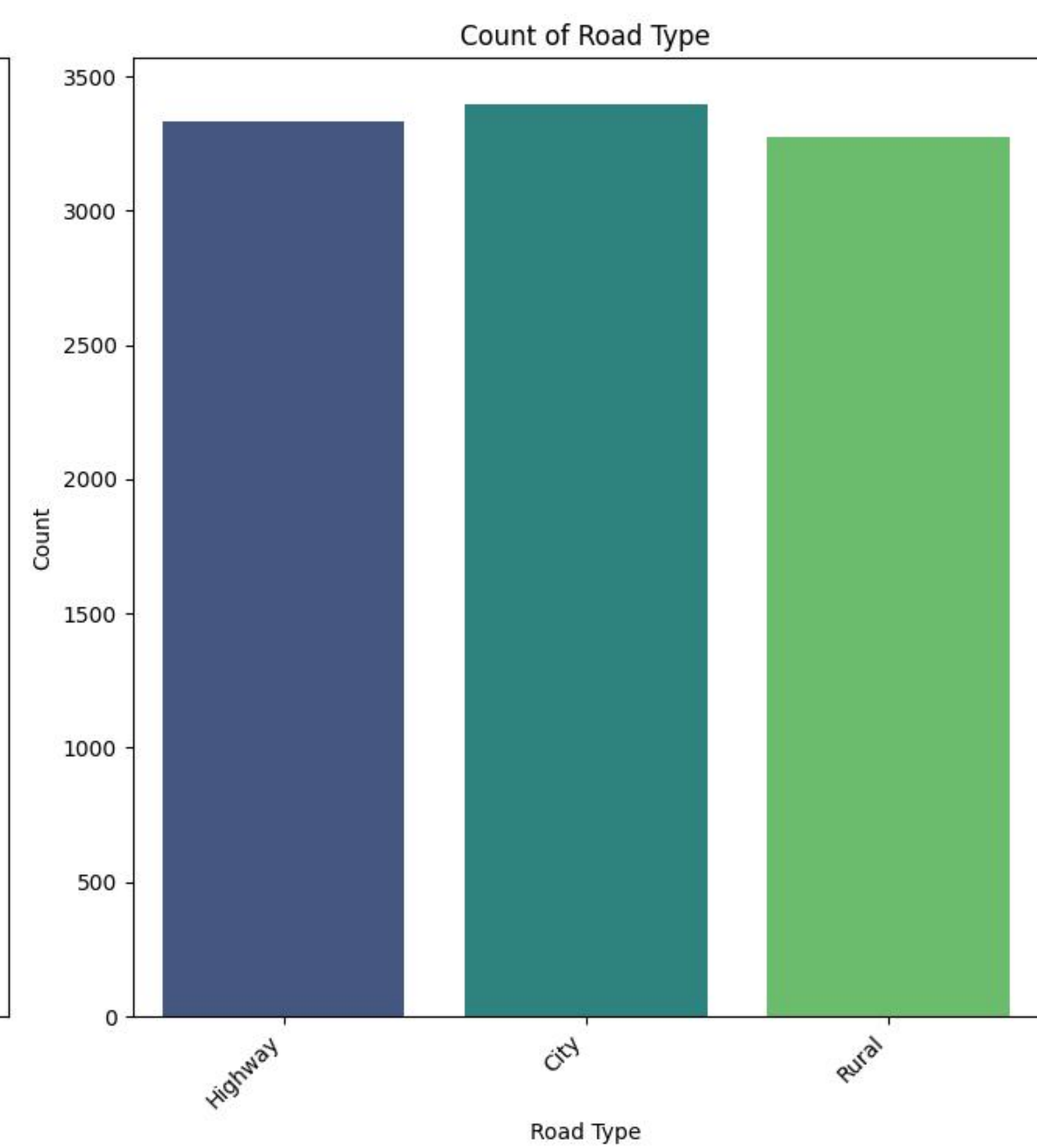
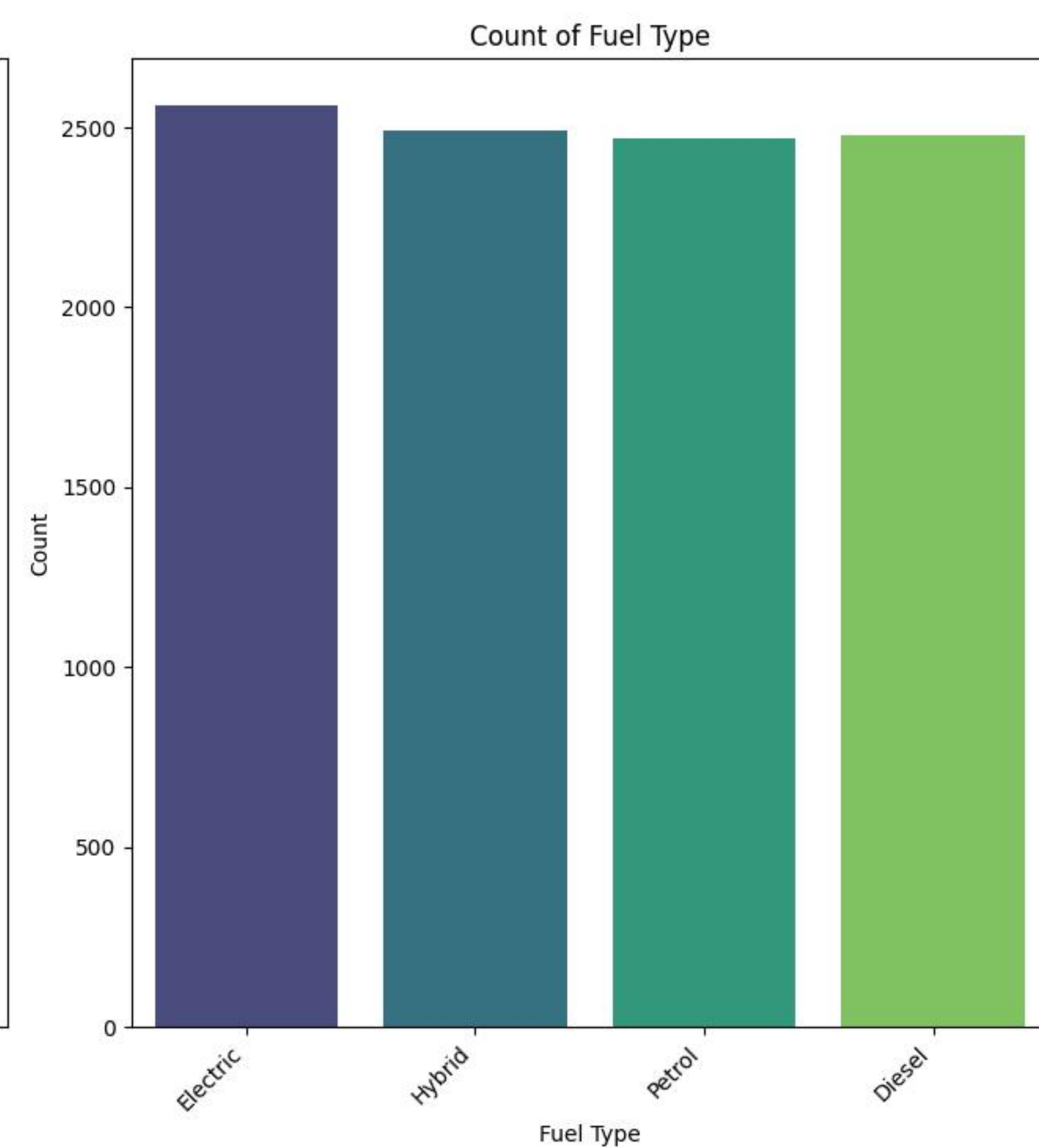
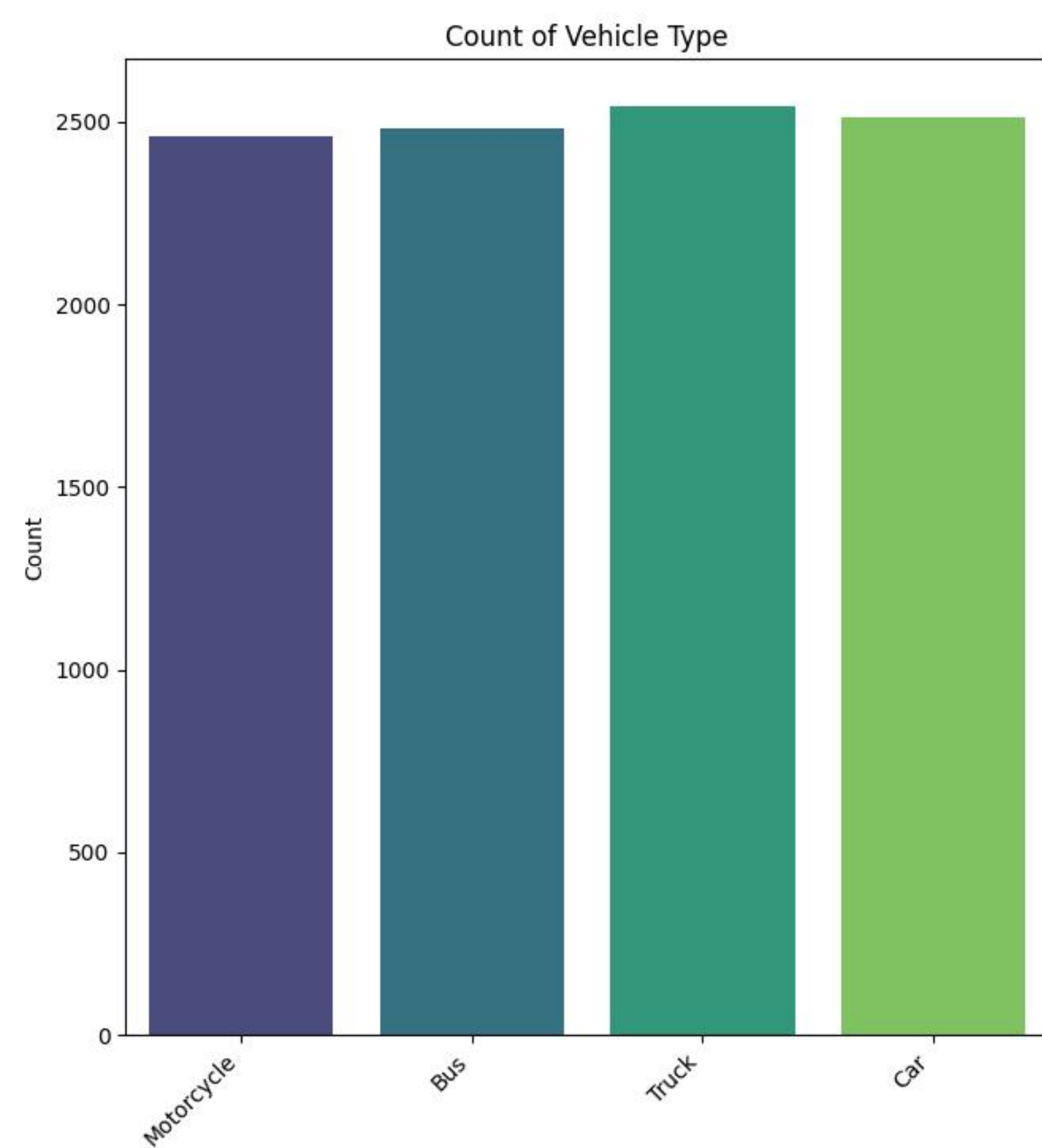
Correlation Matrix of Numerical Features



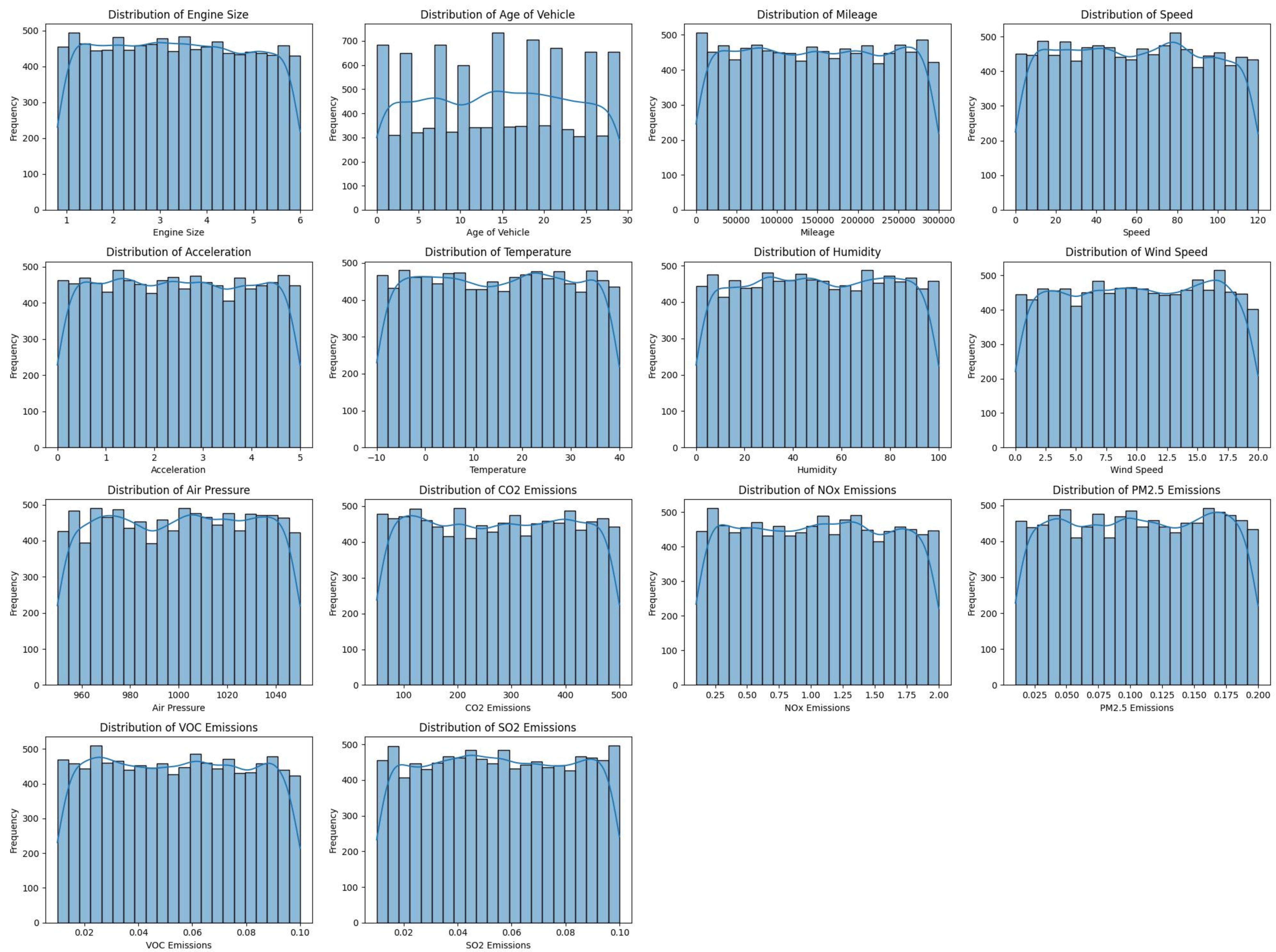


Correlation Matrix of Numerical Features



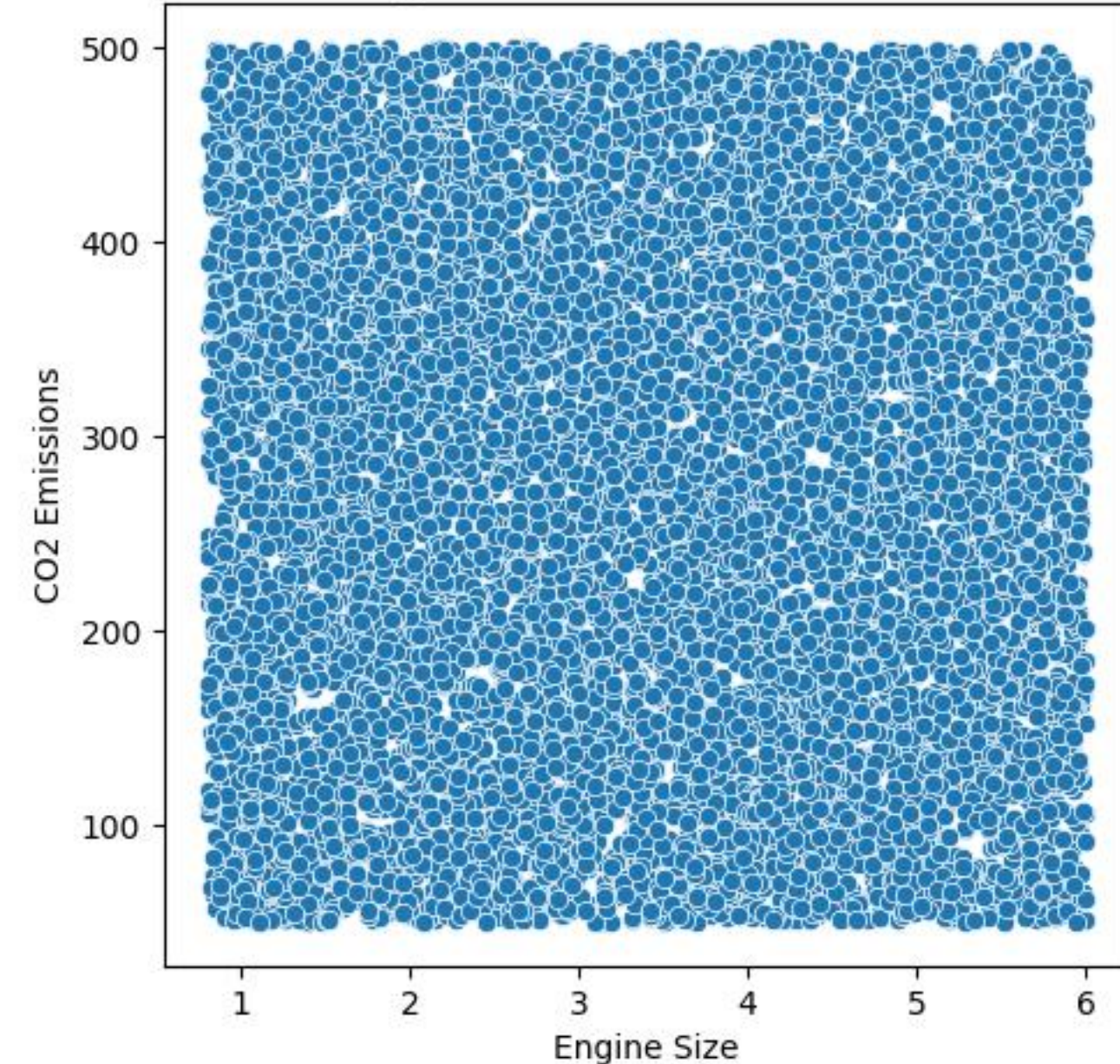




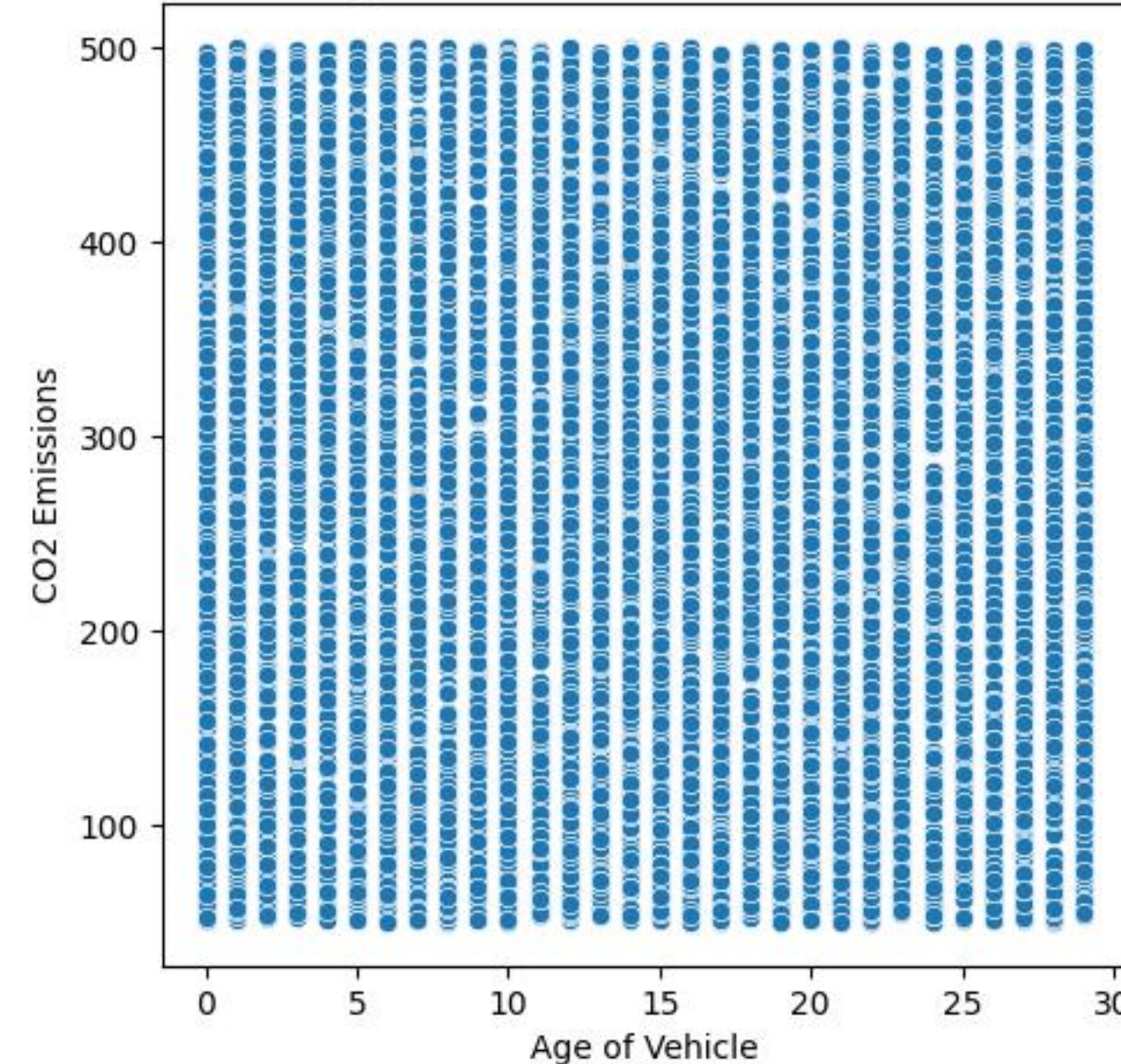




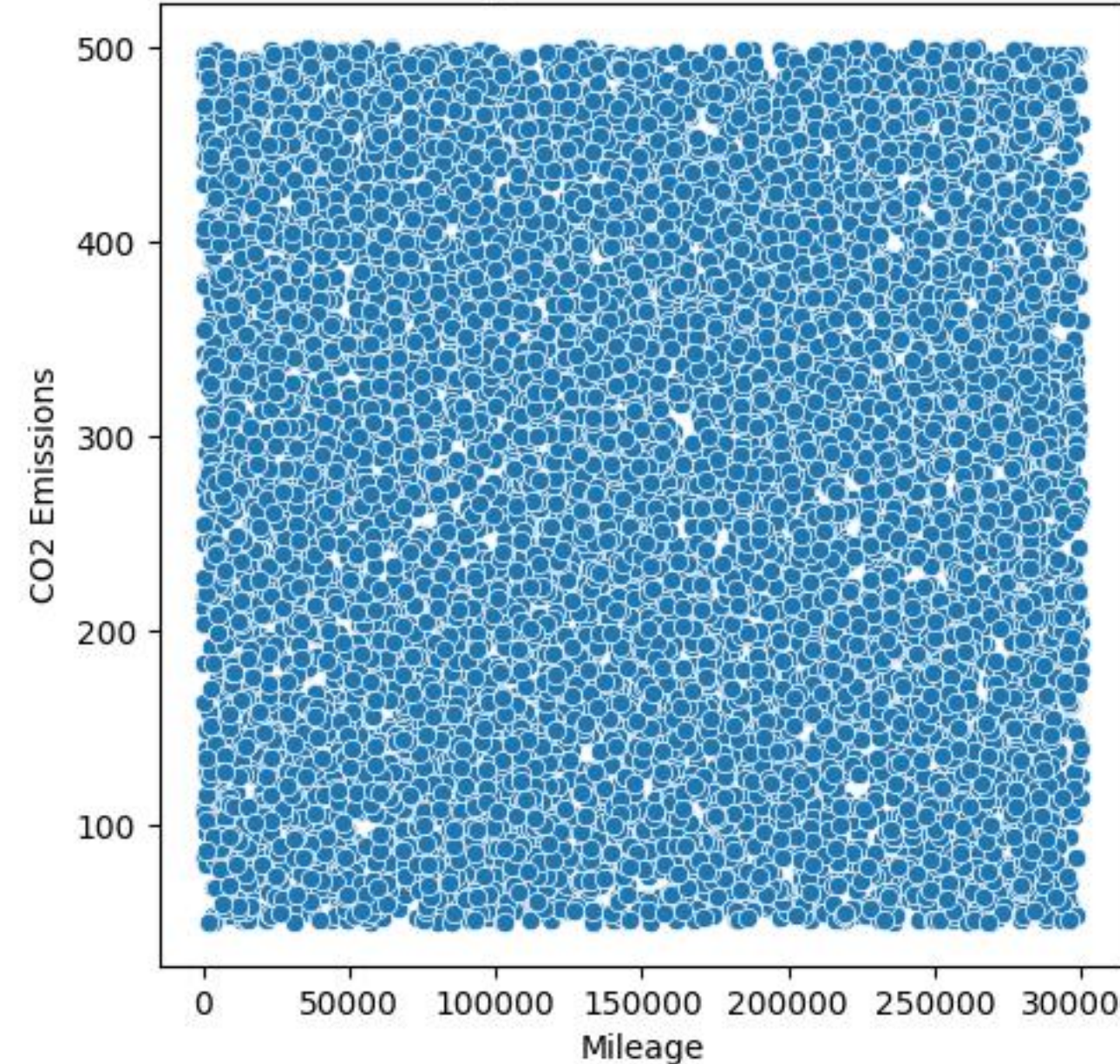
Engine Size vs. CO2 Emissions



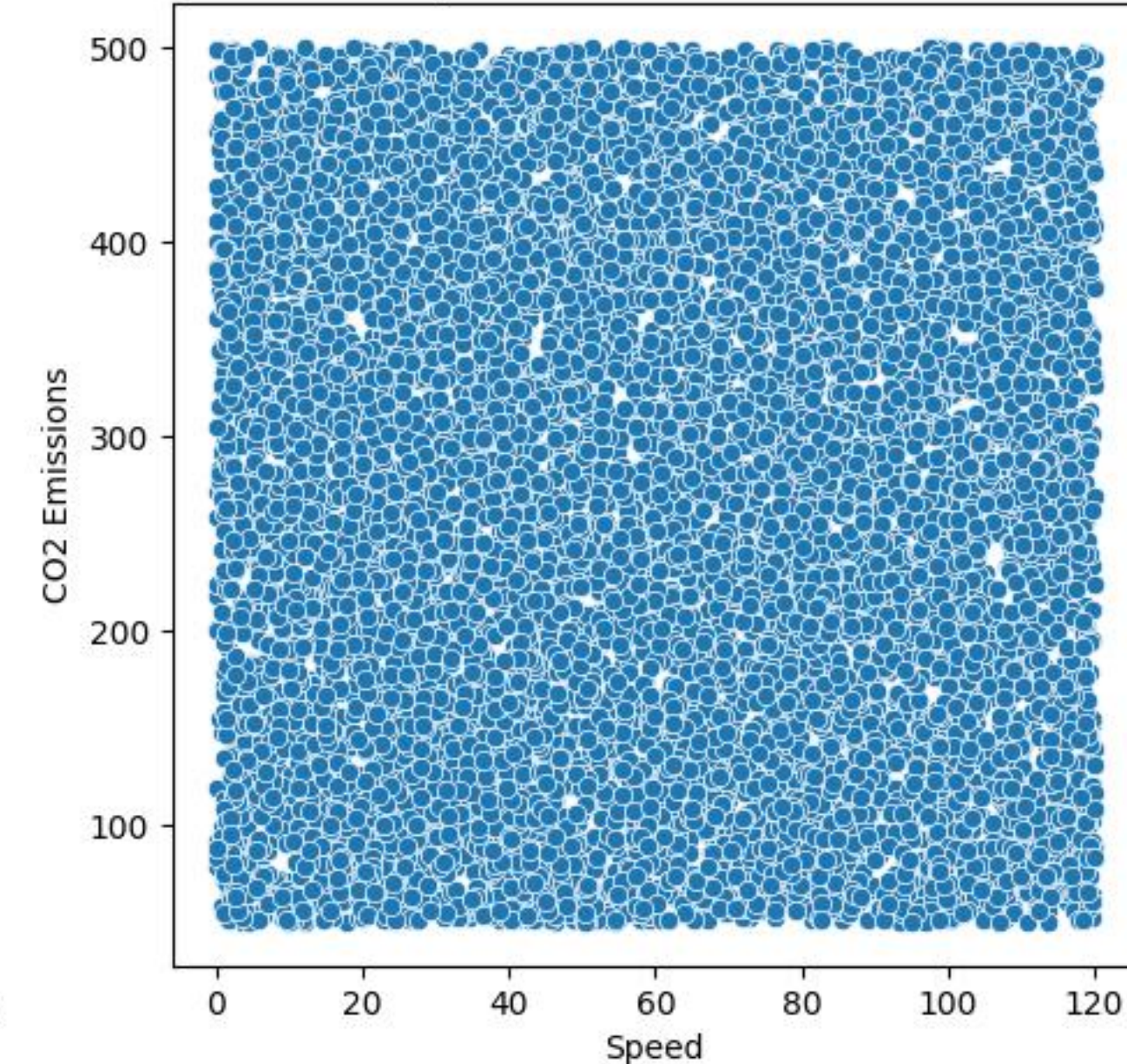
Age of Vehicle vs. CO2 Emissions



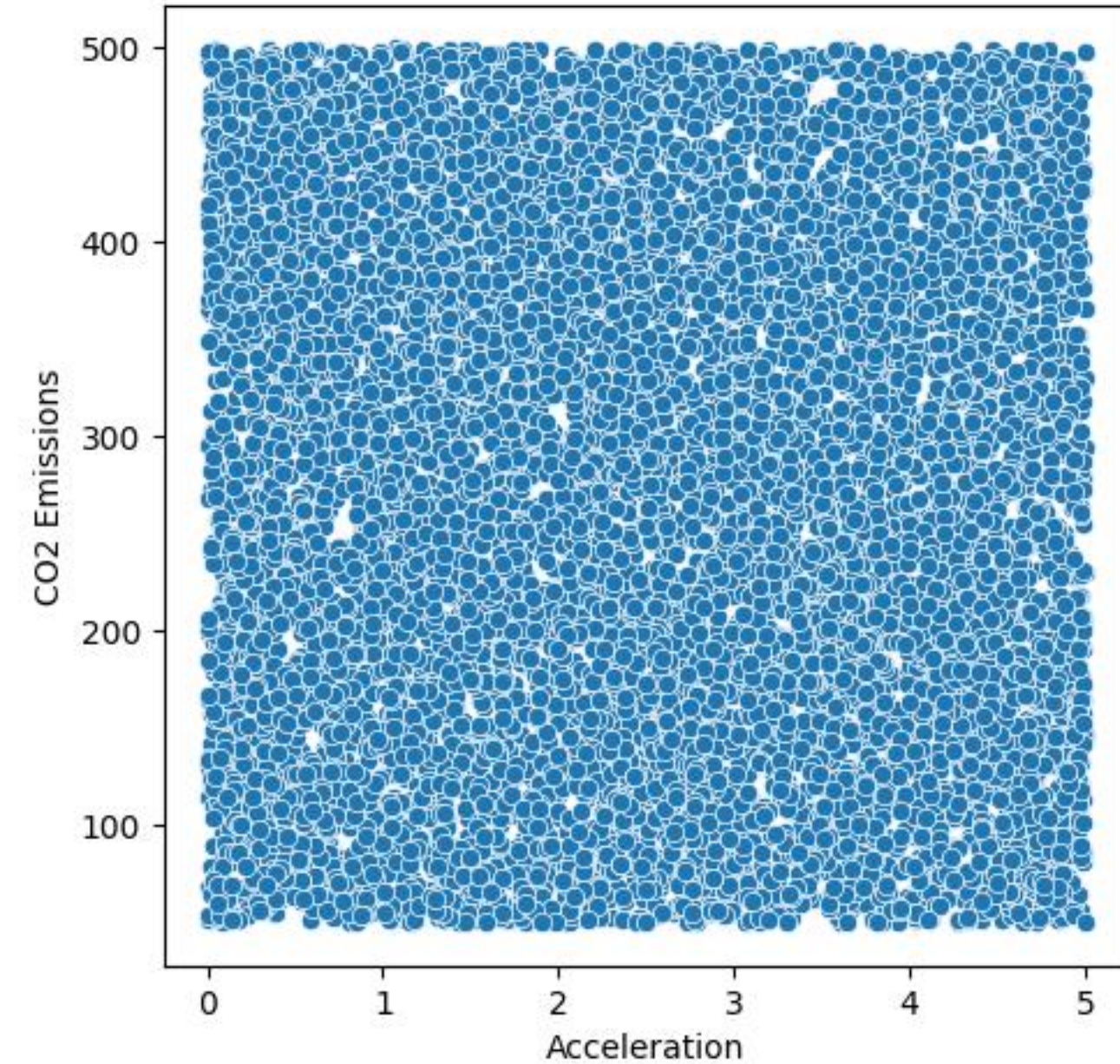
Mileage vs. CO2 Emissions



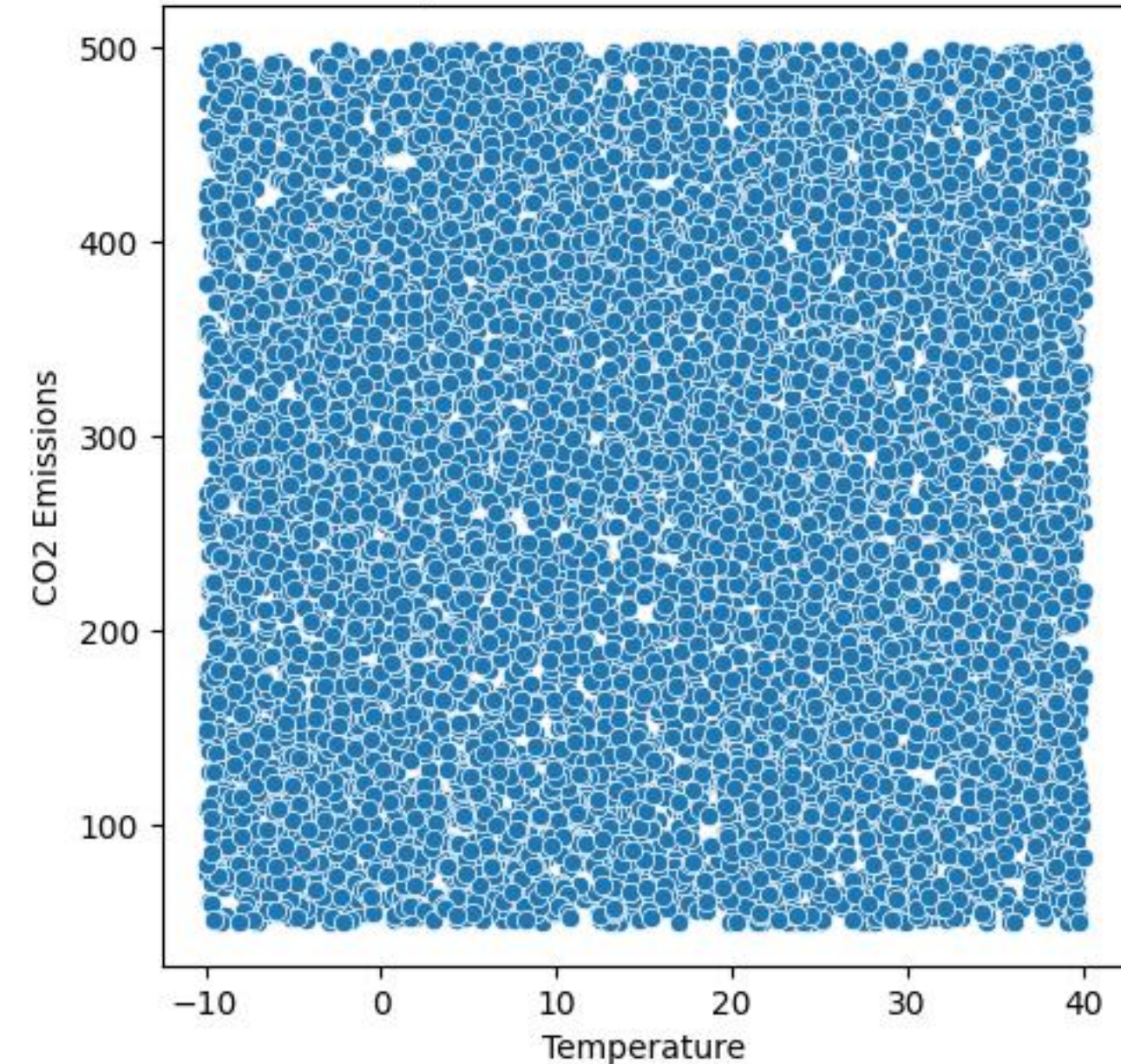
Speed vs. CO2 Emissions



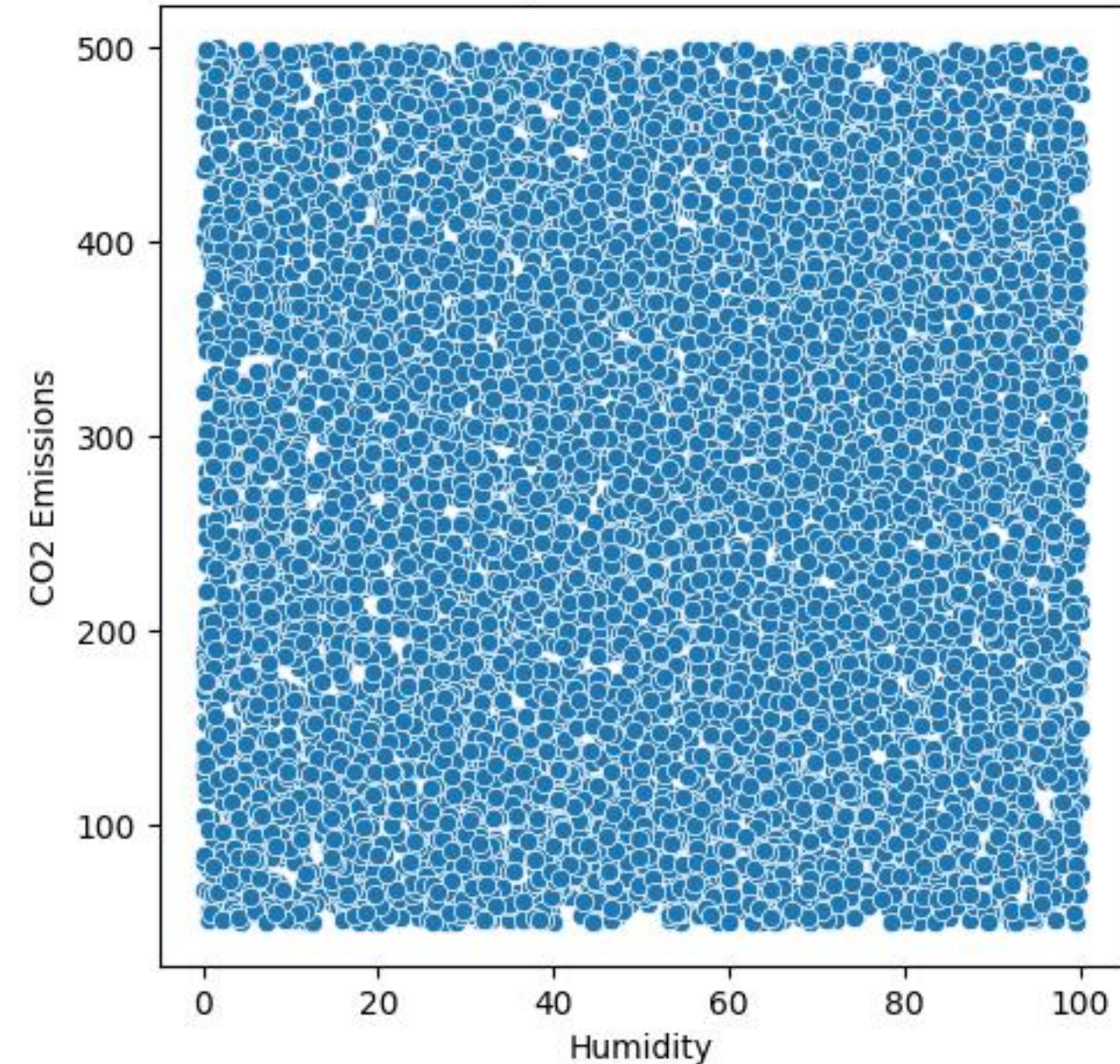
Acceleration vs. CO2 Emissions



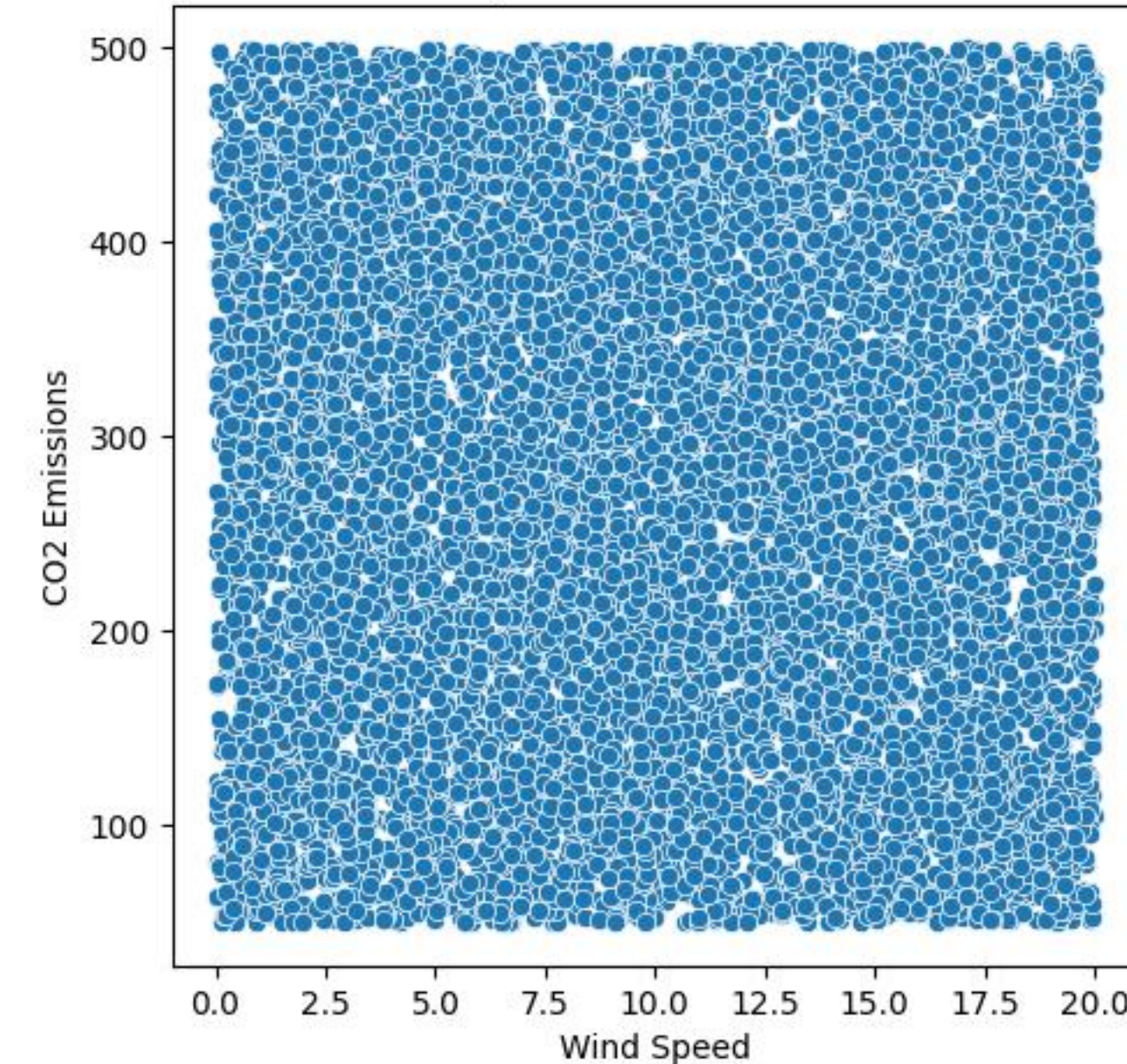
Temperature vs. CO2 Emissions



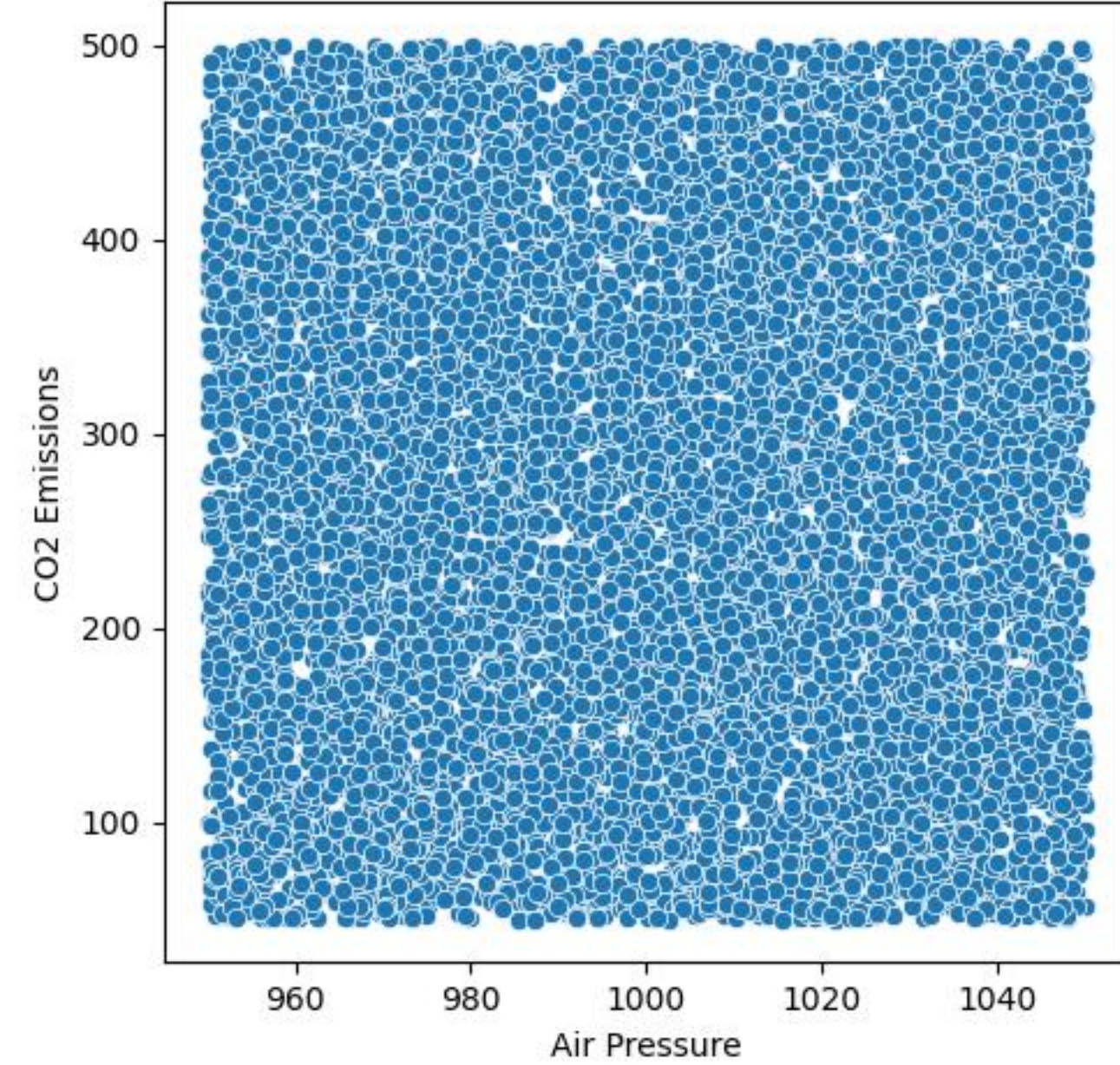
Humidity vs. CO2 Emissions



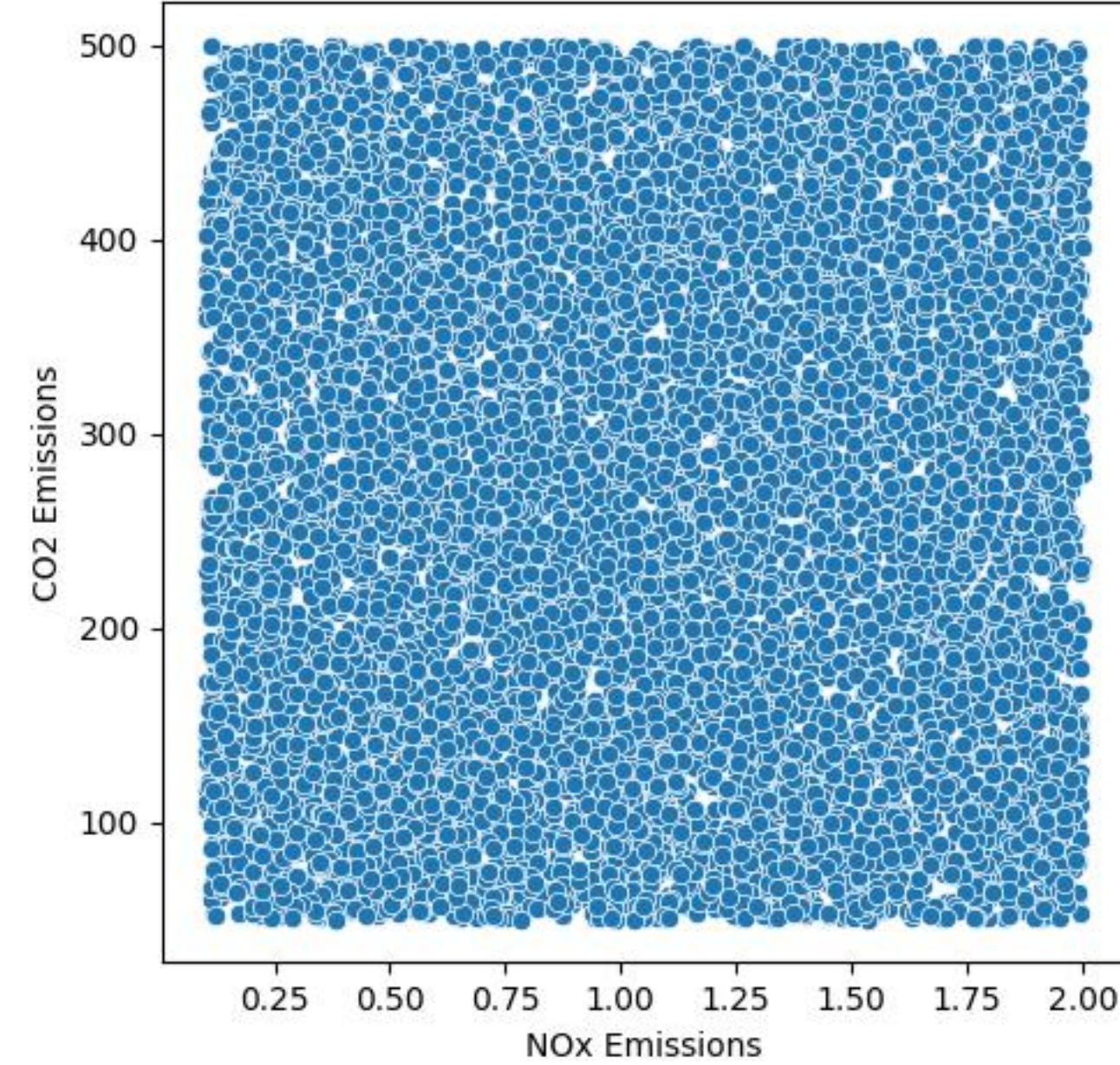
Wind Speed vs. CO2 Emissions



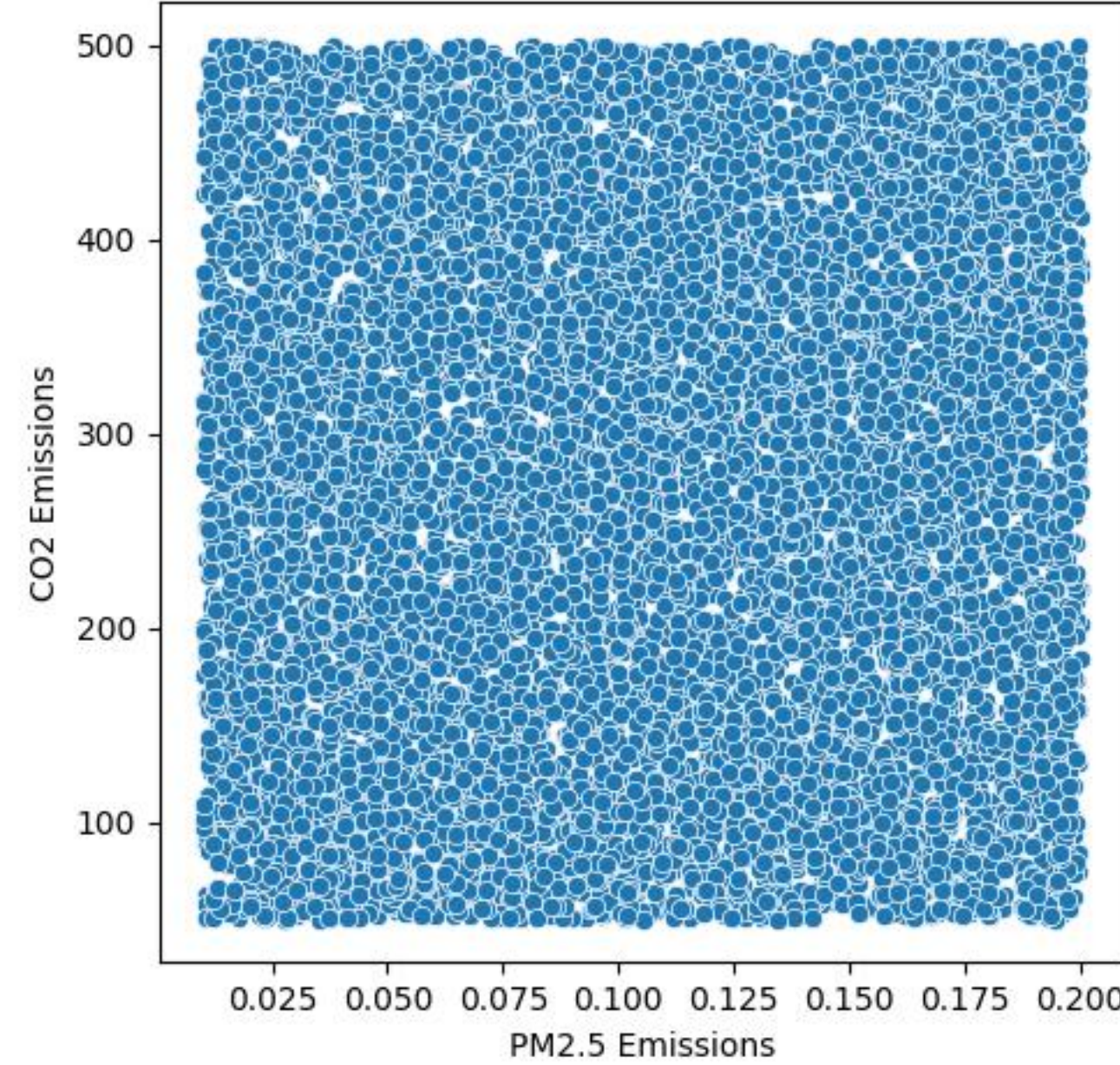
Air Pressure vs. CO2 Emissions



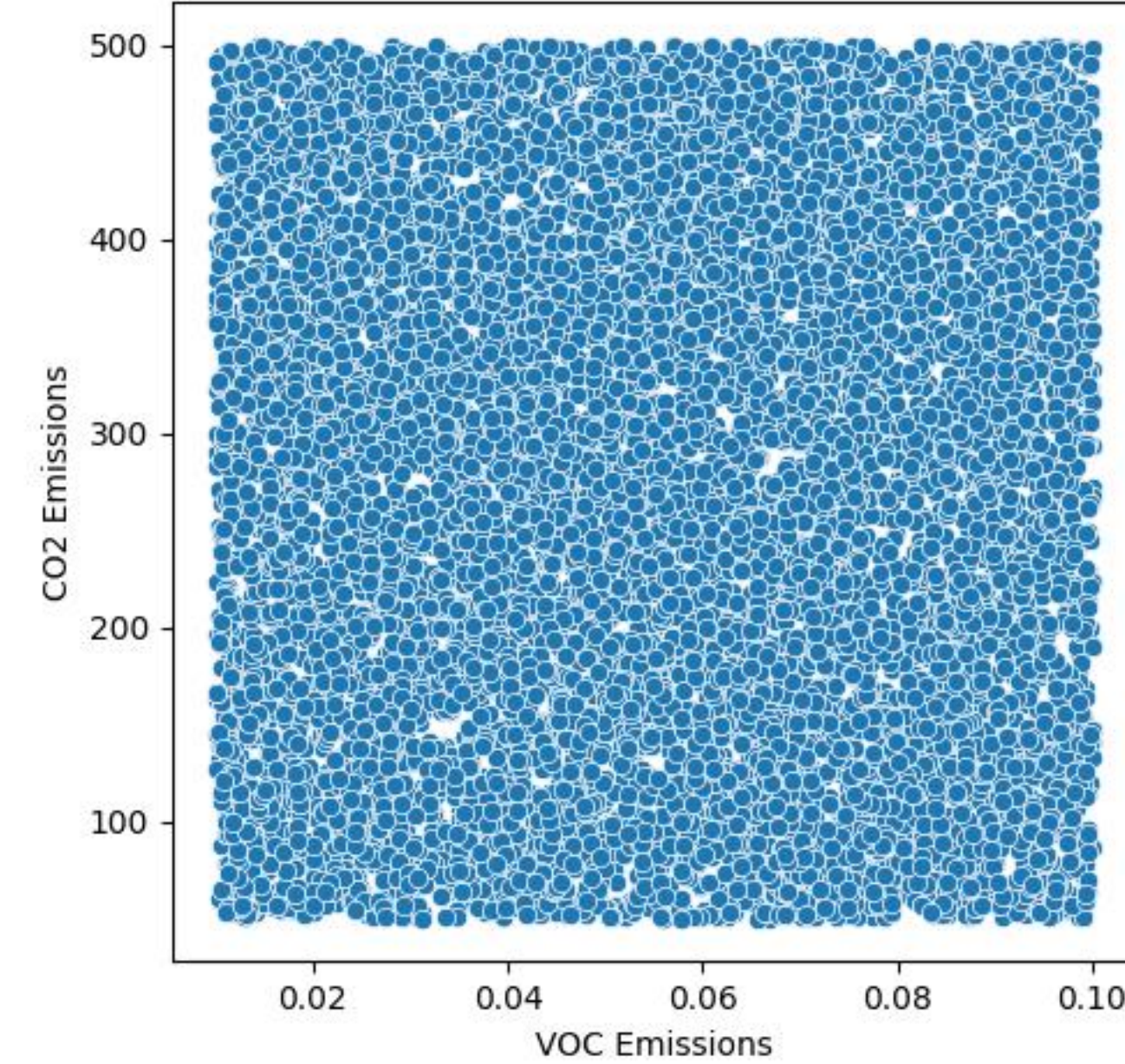
NOx Emissions vs. CO2 Emissions



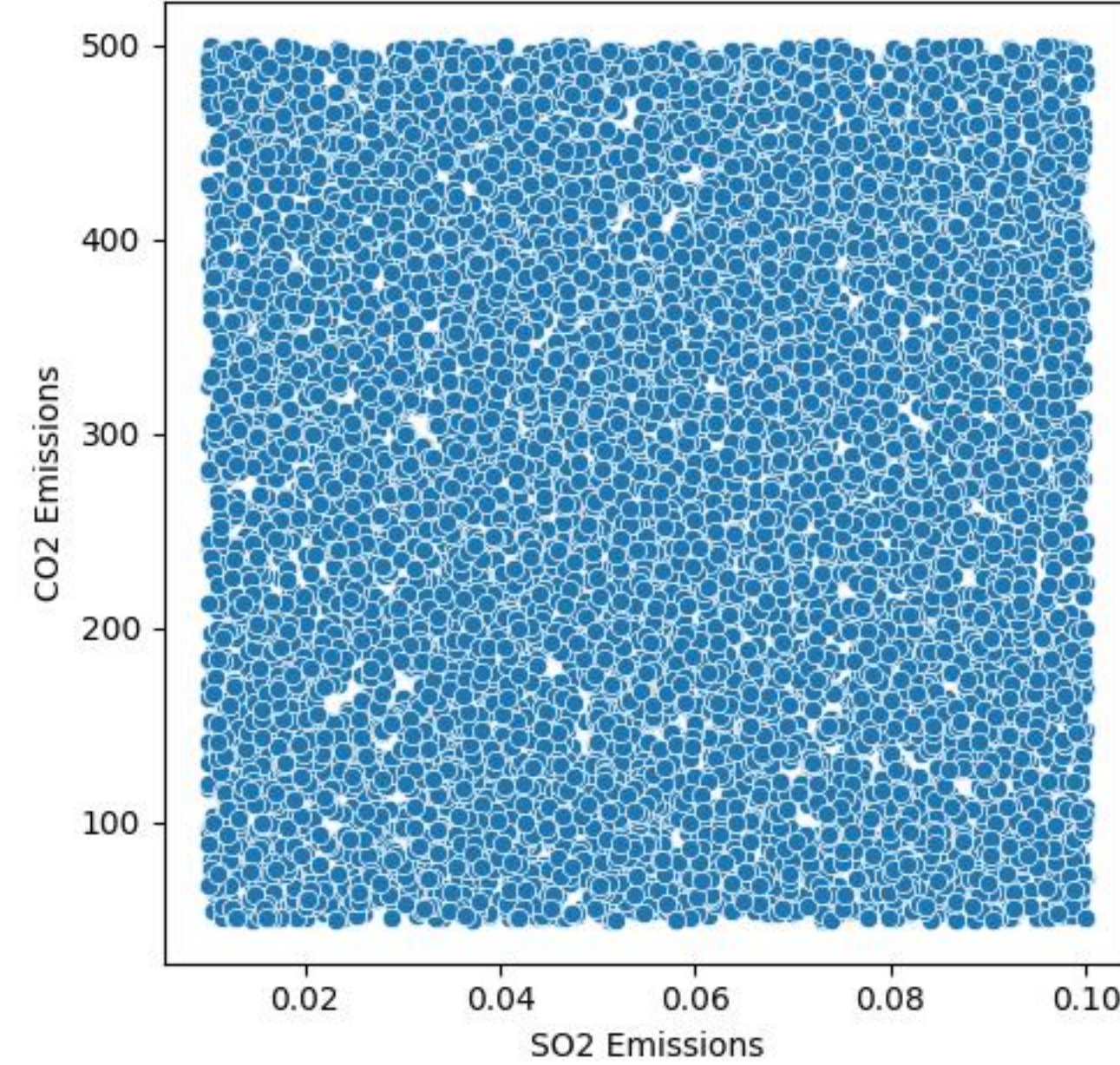
PM2.5 Emissions vs. CO2 Emissions



VOC Emissions vs. CO2 Emissions

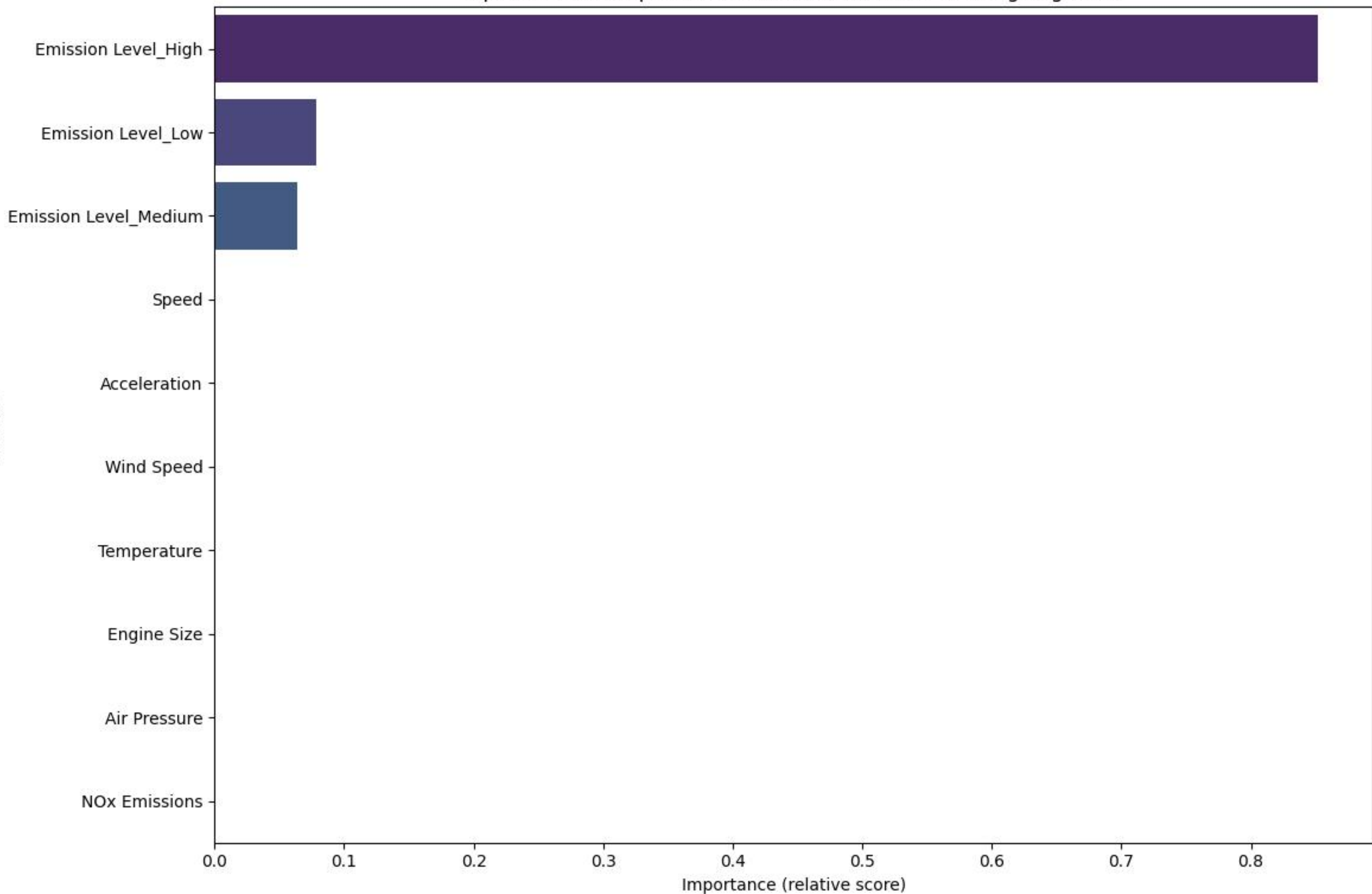


SO2 Emissions vs. CO2 Emissions



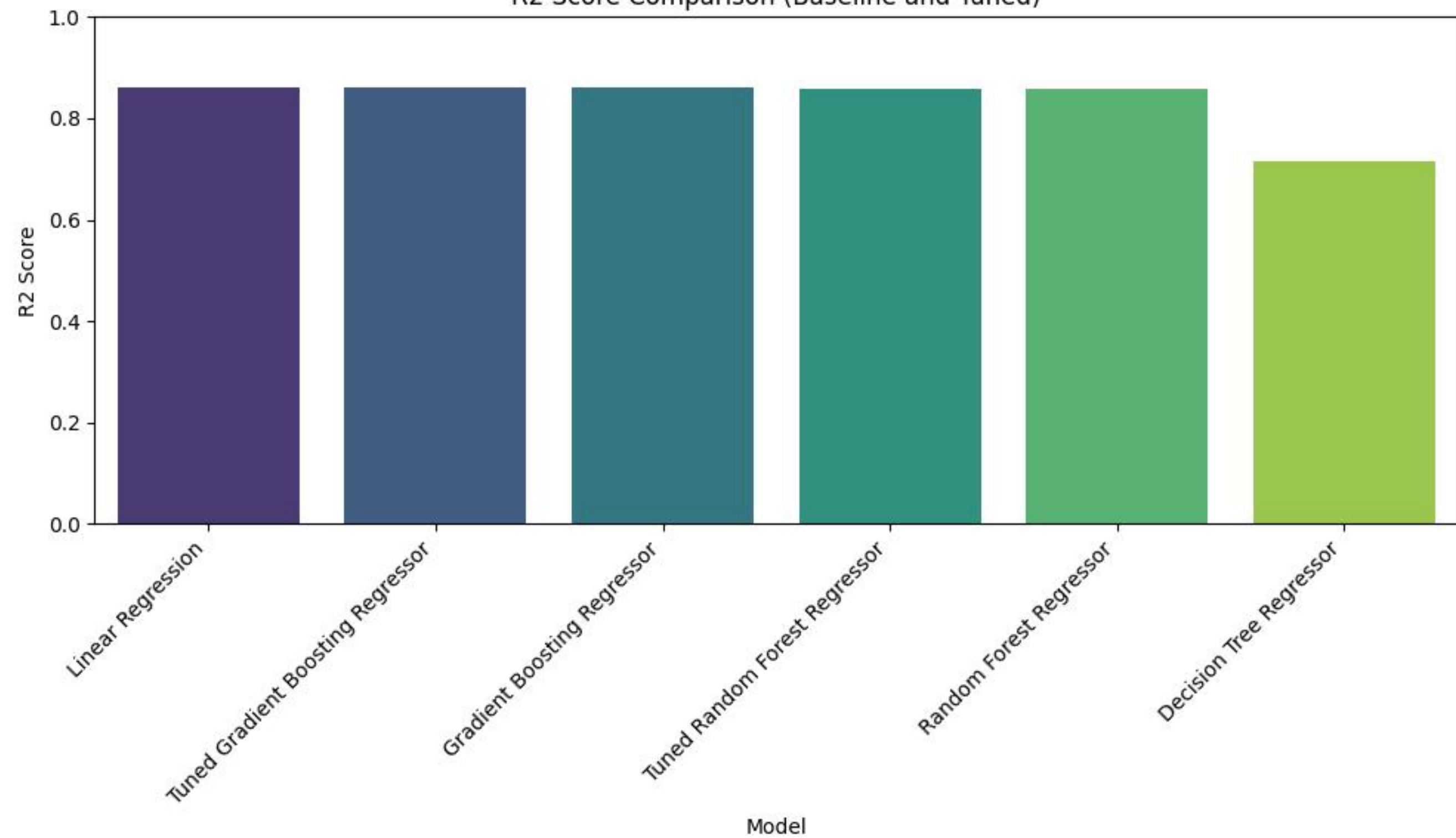


Top 10 Feature Importances for Tuned Gradient Boosting Regressor

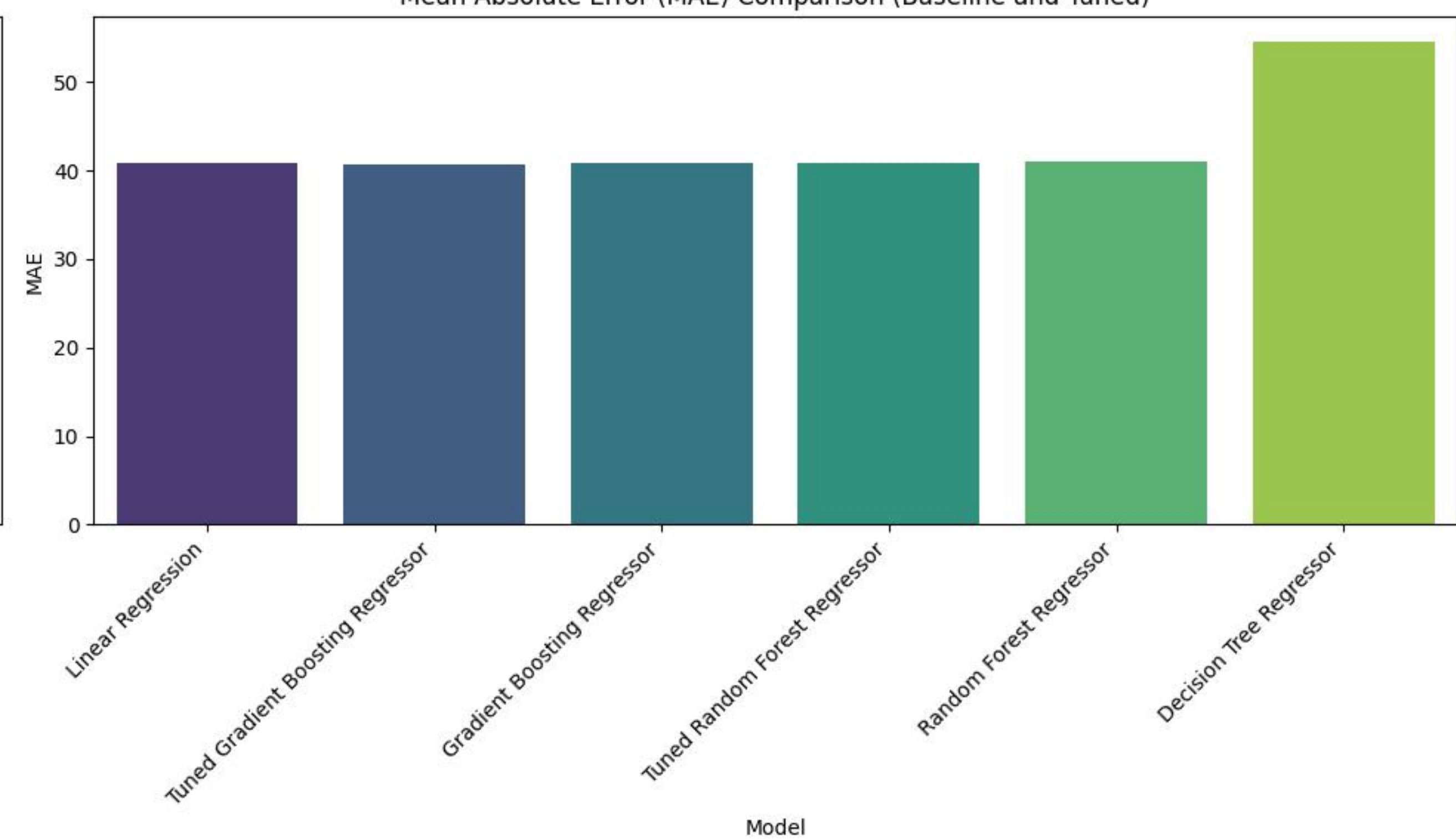




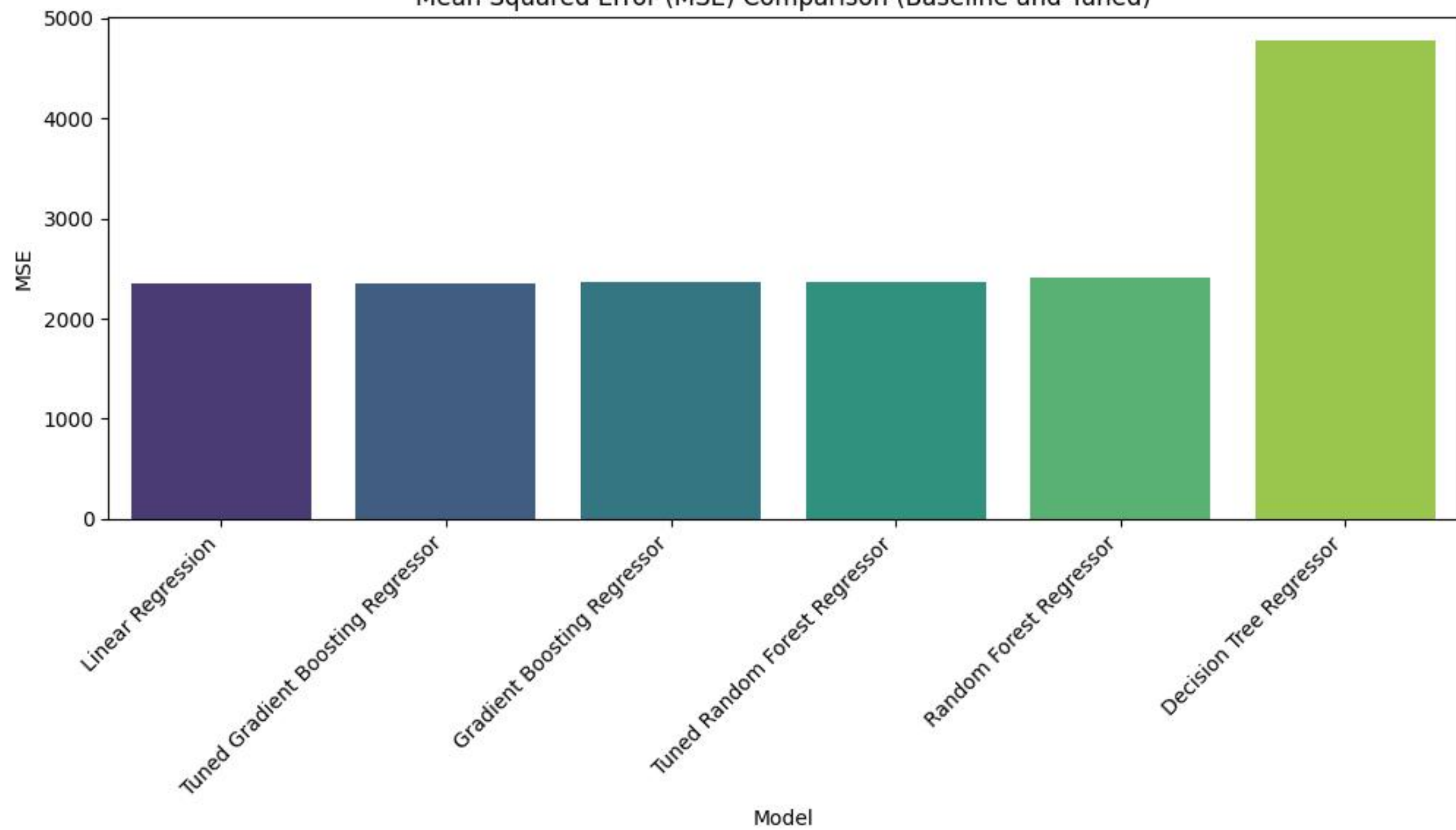
R2 Score Comparison (Baseline and Tuned)



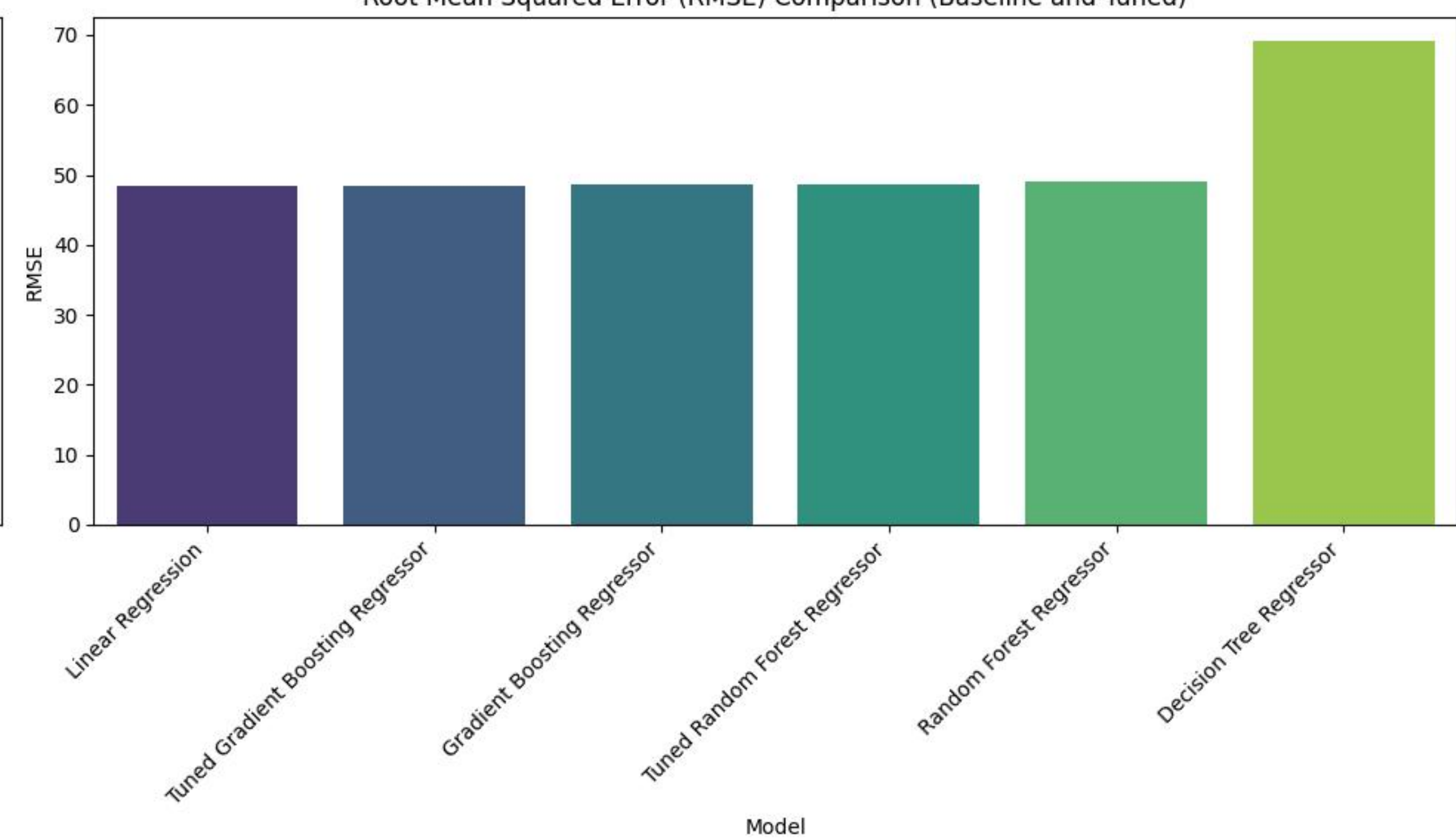
Mean Absolute Error (MAE) Comparison (Baseline and Tuned)



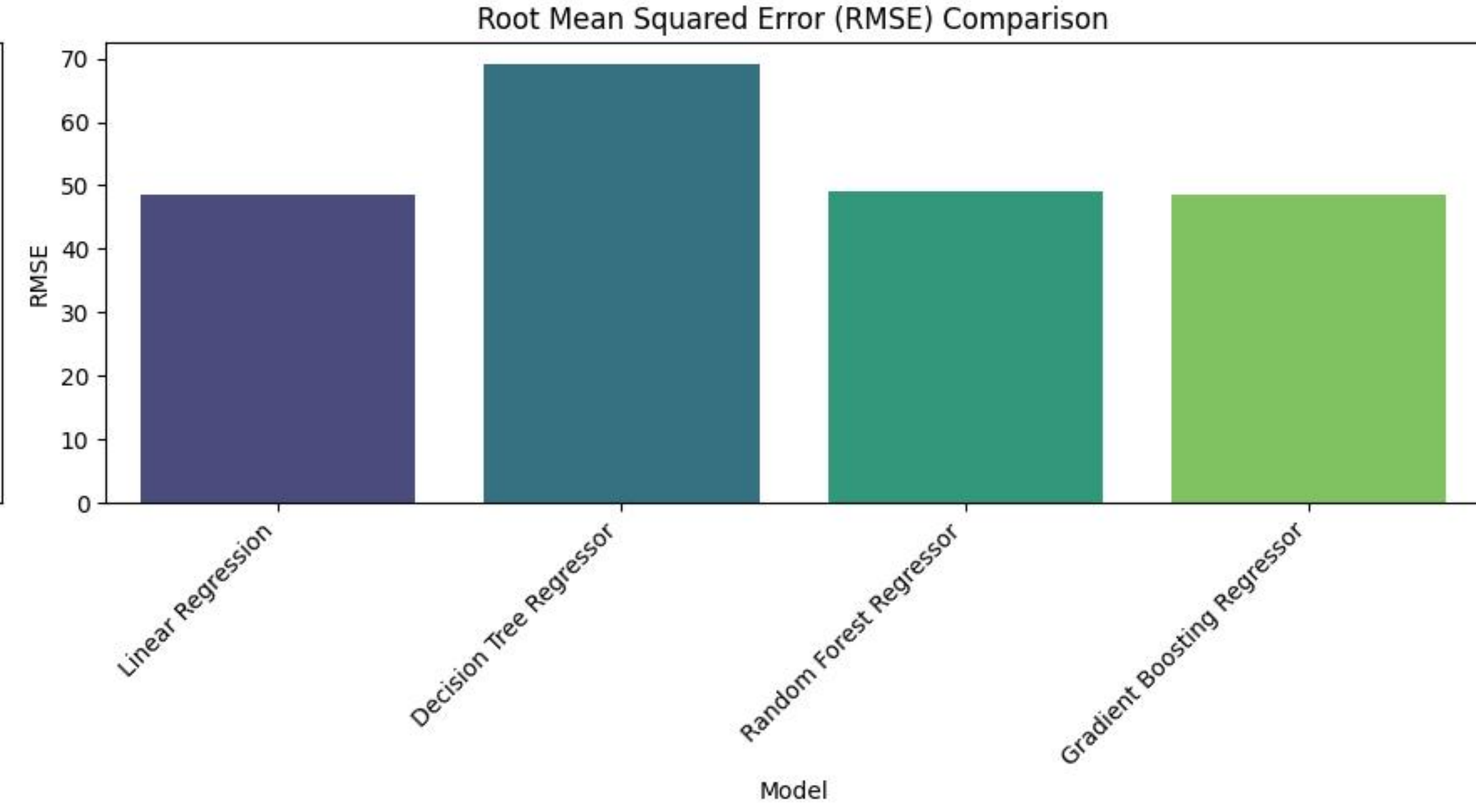
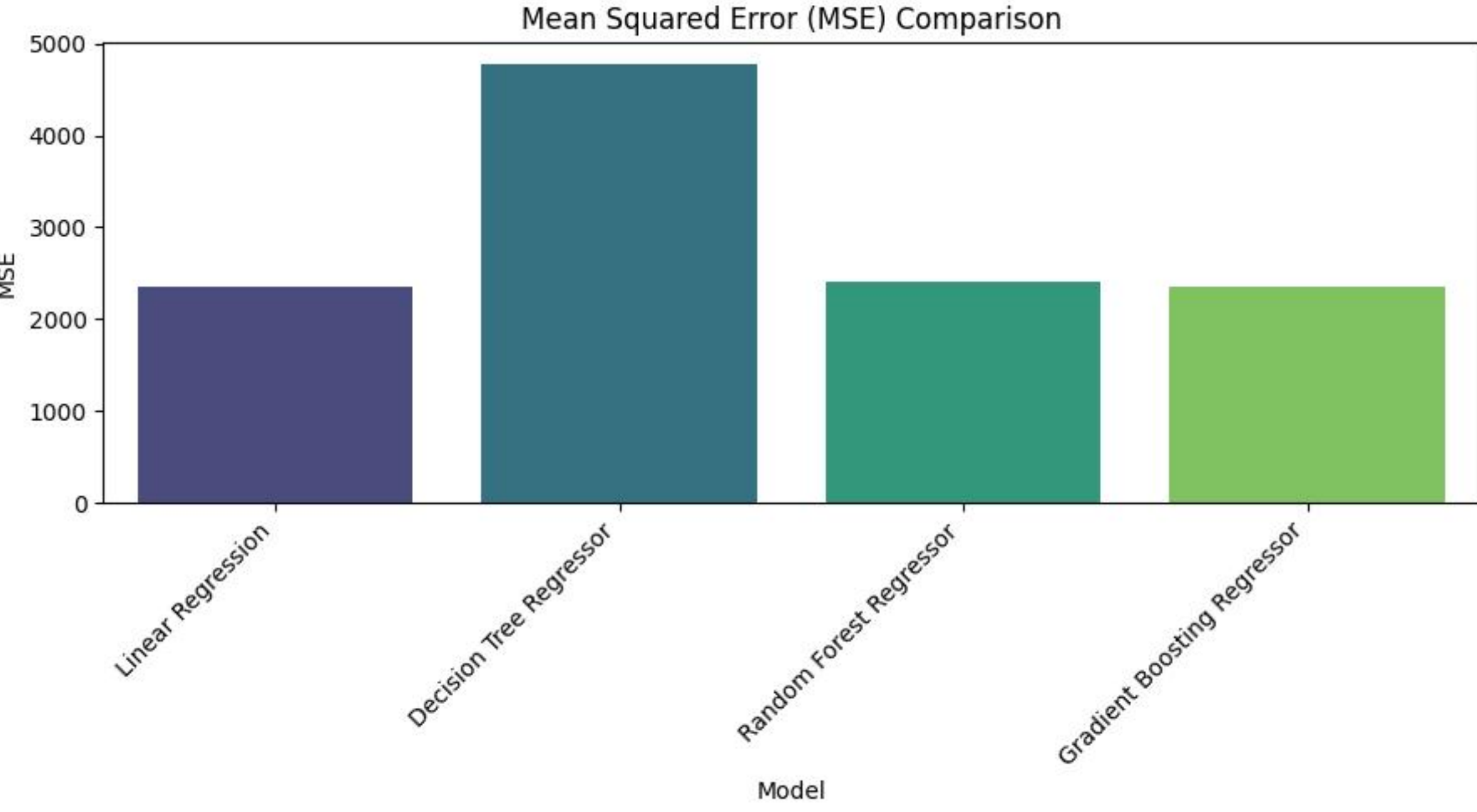
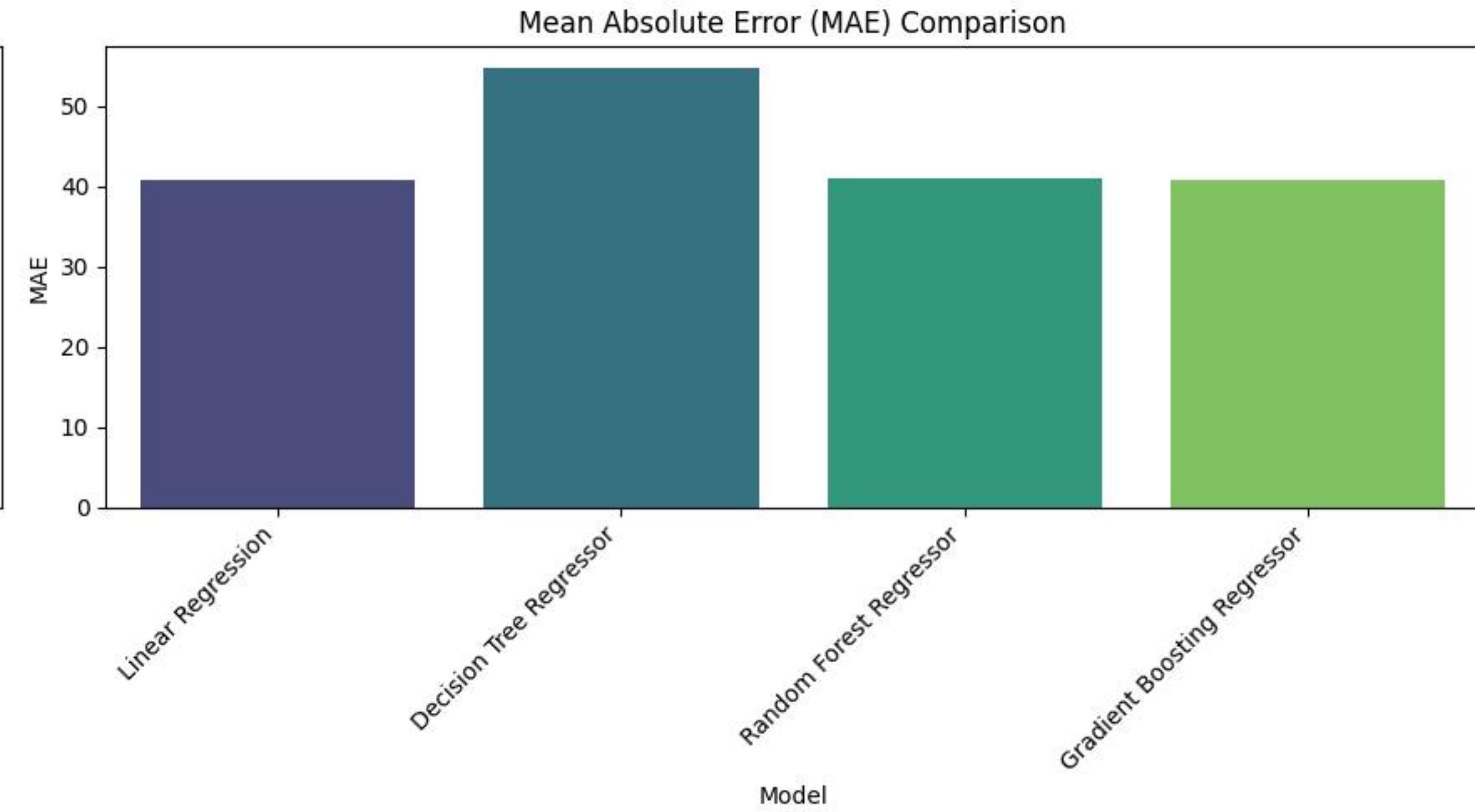
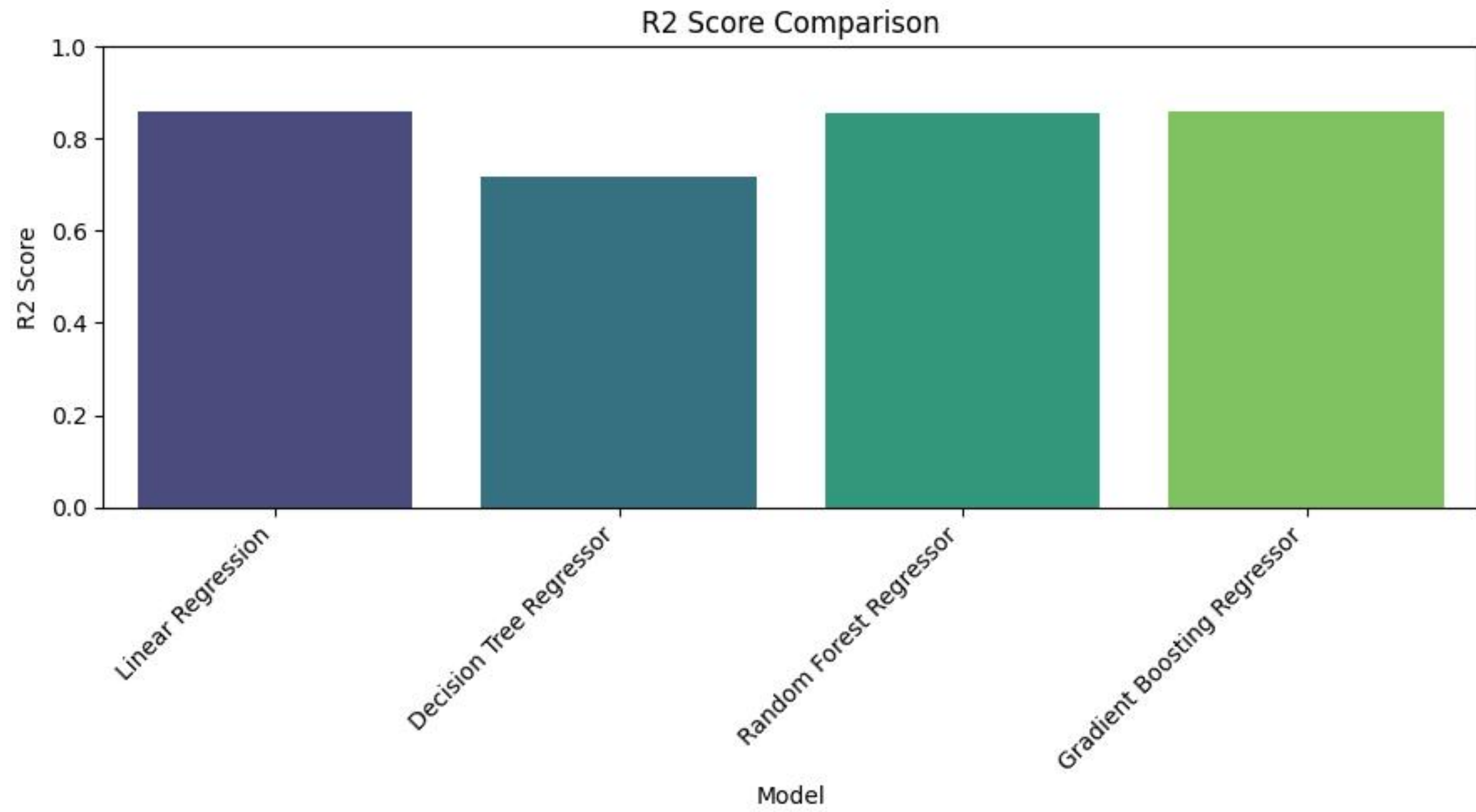
Mean Squared Error (MSE) Comparison (Baseline and Tuned)



Root Mean Squared Error (RMSE) Comparison (Baseline and Tuned)









Residual Plot (Tuned Gradient Boosting Regressor)

